

GRAPH-GUIDED LONG-HORIZON MPC WITH WORLD MODELS

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ABSTRACT

Terminal latent distance can poorly reflect progress in long-horizon world-model planning. We add a temporal distance representation, a graph of offline experience, and a budget-conditioned reachability critic to a frozen LeWorldModel (LeWM). Graph search supplies an intermediate target. The critic scores predicted endpoints by the expected time to reach goal, and the controller switches to latent L2 near the goal. We keep the encoder, predictor, and cross-entropy method (CEM) search settings fixed. We bound the number of replanning calls needed to reach the goal under explicit execution, model, search, and critic error conditions. We also bound critic error across horizons and state when its score equals a truncated expected time to reach the goal. On fixed evaluation tasks, cross-episode success increases from 14.4 to 42.4% on Push-T, 29.2 to 65.6% on Reacher, and 16.4 to 27.6% on Cube. Improvements are largest beyond one planning horizon; short-range losses remain on Push-T and Reacher. Ablations show that graph guidance and the critic are complementary on Push-T, whereas Cube benefits primarily from the learned temporal representation.

1 INTRODUCTION

Joint-embedding world models enable control by predicting the consequences of candidate actions in a learned latent space (Zhou et al., 2024; Maes et al., 2026). Their planning objective often compares a predicted terminal latent with an encoded goal image. This avoids learning a task-specific reward, but latent Euclidean distance need not reflect the time or actions required to reach the goal. On our Push-T tasks, LeWM’s terminal-L2 planner succeeds at 91.2% for goals 25 steps ahead, 40.8% at 50 steps, and 13.2% at 100 steps. Cross-episode success is 14.4%, despite replanning every 25 steps.

We use a graph of offline observations to choose intermediate targets. A temporal distance representation (TDR) organizes frozen latents into a graph. Graph search selects a cluster along a shortest route to the goal using a calibrated lookahead distance. CEM then optimizes a combination of distance to that cluster and a budget-conditioned reachability score. Once the current state is close to the goal in TDR space, the base cost switches to LeWM’s latent L2. The critic remains active. We train the TDR and critic while keeping the dynamics model fixed; CEM supplies the actions.

Graph connectivity does not ensure executable routes: proximity edges may be infeasible, and CEM can miss useful actions. We analyze the normalized subgoal and critic cost and prove finite-time progress and local convergence under explicit execution, model, search, and observability conditions. We also explain why optimizing reachability separately at each horizon need not describe the time to reach the goal under any single policy.

The same configuration improves over LeWM at offsets 50 and 100 and on cross-episode goals in all three environments. On Cube, however, TDR alone achieves higher cross-episode success than the full method. Our main contributions are:

1. A graph-guided objective for latent model predictive control (MPC), combining dataset subgoals, a remaining-time budget, and a local L2 switch (section 3).
2. A method-specific progress and convergence analysis, including the effect of CEM approximation and population normalization (section 4).
3. A controlled comparison on three environments, with component ablations and diagnostics that distinguish long-range guidance from local execution (section 5).

2 RELATED WORK

Costs and hierarchies for latent planning. LeWM and DINO-WM use predictive latent models for visual planning (Maes et al., 2026; Zhou et al., 2024). TRM learns a temporal pairwise cost on a fixed representation (Li et al., 2026a); RC-aux adds reachability supervision during world-model training (Li et al., 2026b). SAGE uses learned subgoals and action proposals, while Hi-LeWM studies how high-level search interacts with a frozen low-level model (Cheng et al., 2026; Caselli et al., 2026). We choose targets from clusters of dataset observations and keep CEM’s action initialization unchanged in the main configuration.

Graph planning and temporal representations. SPTM, SoRB, and SGM combine stored experience with learned local controllers (Savinov et al., 2018; Eysenbach et al., 2019; Emmons et al., 2020). GAS uses a temporal graph to guide a learned policy (Baek et al., 2025); SSE addresses subgoal feasibility through replay of execution outcomes (Hwang et al., 2026). We adapt GAS’s graph construction to frozen-model MPC. PcLast and ALPS already combine cluster-based planning and CEM; ALPS also learns a behavior prior (Koul et al., 2024; Shehmar et al., 2026). Our contribution combines a frozen pixel world model, a survival cost that uses the remaining budget, and a local switch. We prove a progress bound for their joint objective. HILP proves goal-reaching results under representation and directional-control errors (Park et al., 2024); quasimetric learning treats the asymmetry of temporal distance explicitly (Wang et al., 2023). We analyze how the controller attaches to the graph, executes each full plan, and normalizes the combined cost.

Values and convergence in MPC. POLO and TD-MPC use learned terminal values to complement short model rollouts (Lowrey et al., 2019; Hansen et al., 2022; 2024). Classical MPC stability depends on suitable terminal-region and local-control conditions (Mayne et al., 2000); stochastic shortest-path theory similarly requires conditions ensuring eventual goal arrival (Bertsekas & Tsitsiklis, 1991). Our guarantees likewise require explicit control and goal-reaching assumptions. We bound graph progress and error propagation; the neural training loss alone does not establish convergence.

3 METHOD

Figure 1 summarizes the offline preparation and online planning procedure.

Frozen model and time units. The encoder E maps a 224×224 image to a 192-dimensional latent. The LeWM predictor P uses a three-block history. One action block contains $\Delta = 5$ environment actions; CEM plans $K = 5$ blocks. The controller executes $L = K\Delta = 25$ steps before replanning, unless it reaches the goal first. Every main comparison uses 300 candidates, 30 iterations, and 30 elites. LeWM’s base cost is $c_{L2}(\hat{z}_K) = \|\hat{z}_K - z_g\|^2$. We change the scoring function while retaining the encoder, predictor, and search settings. All distances without a squared norm below are unsquared Euclidean distances.

3.1 TEMPORAL REPRESENTATION AND GRAPH

Write $d_\psi(z, z') = \|\psi(z) - \psi(z')\|$, with $\psi : \mathbb{R}^{192} \rightarrow \mathbb{R}^{32}$. Following HILP and GAS (Park et al., 2024; Baek et al., 2025), two networks learn the negative-distance value $U(z, z_g) = -d_\psi(z, z_g)$ using expectile TD. For a nonterminal transition, the target is $q = -1 + \gamma \min_{i=1,2} U_i^-(z', z_g)$, with $\gamma = 0.99$ and EMA targets:

$$\mathcal{L}_{\text{TDR}} = \mathbb{E}[\ell_{0.99}(q - U(z, z_g))], \quad \ell_\tau(u) = |\tau - \mathbf{1}_{\{u < 0\}}|u^2. \quad (1)$$

We set the goal’s terminal value to zero. With probability 0.625, we sample a future goal frame at a geometric offset with parameter $1 - \gamma$; otherwise, we sample uniformly from the dataset. The learned distance approximates discounted temporal structure symmetrically. We calibrate its scale using $u_k = \text{median}_t d_\psi(z_t, z_{t+k})$ on training trajectories.

Graph construction uses radius $r_{\text{TD}} = u_{12}$ in TDR units and a separate integer temporal lag k_{TE} in environment steps. Let F_t be the first later frame in the same episode at TDR distance at least

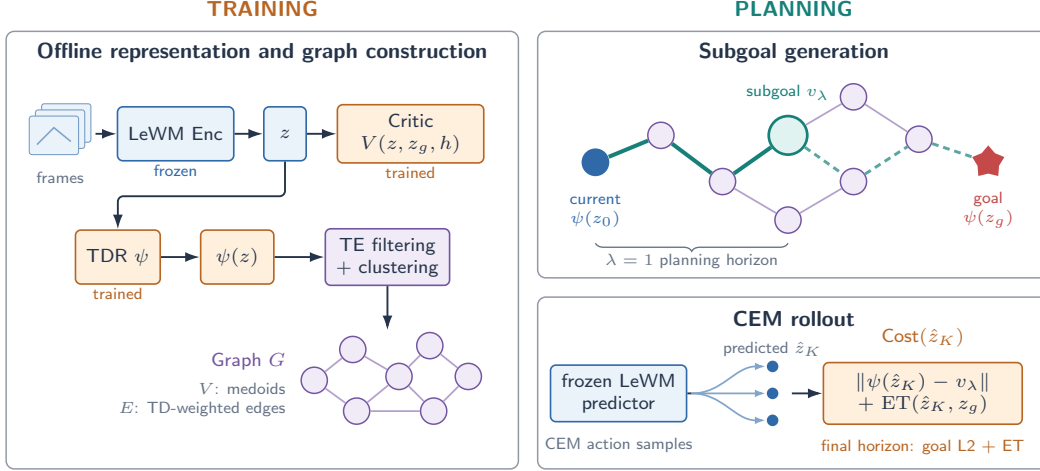


Figure 1: Graph-guided latent MPC. Offline, the frozen encoder supplies latents for the TDR, graph, and reachability critic; critic labels are derived from the TDR, so no stage reads simulator state. Online, graph search chooses a target and the frozen predictor rolls out CEM candidates. CEM ranks endpoints by combining target distance and the survival score after normalizing each over its candidate population. The controller switches the base cost to goal L2 near the goal. In the schematic, v_λ denotes the subgoal coordinate y_q ; edge weights use TDR distance.

r_{TD} , or the last frame if no such crossing exists. The temporal-efficiency filter retains frames whose nonzero displacement vectors satisfy

$$TE(z_t) = \frac{\langle \psi(F_t) - \psi(z_t), \psi(z_{t+k_{TE}}) - \psi(z_t) \rangle}{\|\psi(F_t) - \psi(z_t)\| \|\psi(z_{t+k_{TE}}) - \psi(z_t)\|} \geq 0.9. \quad (2)$$

We clip the lag at the episode end and exclude comparisons with zero displacement. We cluster retained embeddings sequentially, admitting points within $r_{TD}/2$ of a center and updating centers to member means. Node v stores a TDR center $y_v \in \mathbb{R}^{32}$ and a medoid latent $z_v \in \mathbb{R}^{192}$. We link centers at distance at most r_{TD} with weight $\|y_v - y_w\|$. These links suggest local connections but do not ensure that the dynamics can execute them.

For a goal, a temporary node at $\psi(z_g)$ connects to centers within $r_g = \max(r_{TD}, 1.2d_{\min})$, where $d_{\min} = \min_v \|\psi(z_g) - y_v\|$. Dijkstra’s algorithm computes $D_g[v]$, the shortest-path length to this node. The resulting graphs have approximately 20.5k, 535, and 16k nodes on Push-T, Reacher, and Cube, respectively. We keep the graph and its distances fixed during each episode.

3.2 HORIZON-MATCHED SUBGOAL AND LOCAL SWITCH

At a planning call, restrict attachment to $A(z_0) = \{v : \|\psi(z_0) - y_v\| \leq r_{TD}, D_g[v] < \infty\}$ and choose

$$v^* \in \arg \min_{v \in A(z_0)} \{\|\psi(z_0) - y_v\| + D_g[v]\}. \quad (3)$$

Starting with this attachment segment, follow a shortest path and select the first node whose cumulative path length reaches $\lambda = u_{25}$, or the goal if the path ends earlier. Denote its coordinate by y_q . Selecting the first crossing can overshoot λ by one edge. The theory assumes that an action sequence can reach near this target; median calibration alone does not establish that assumption. The base cost is

$$b(\hat{z}_K) = \begin{cases} \|\hat{z}_K - z_g\|^2, & d_\psi(z_0, z_g) \leq \lambda, \\ \|\psi(\hat{z}_K) - y_q\|, & d_\psi(z_0, z_g) > \lambda. \end{cases} \quad (4)$$

The controller recomputes the switching decision at each call. The graph branch requires $A(z_0) \neq \emptyset$; disconnected or unattached states have no finite route to follow. The controller uses the nearest node as a fallback in this case, but the graph-progress theorem does not cover that fallback.

3.3 REACHABILITY SCORE AND THE REMAINING BUDGET

We query the five-member critic ensemble $V(z, z_g, h) \in [0, 1]$ at $h = 0, 5, \dots, 50$. We train it with binary cross-entropy on hindsight first-hit labels $\mathbf{1}_{\{\tau \leq h\}}$, cross-episode negatives, and predicted latents under logged actions. Here τ is the first logged step with $d_\psi(z_{t+k}, z_g) < u_2$. Cross-episode pairs receive label 0 only when $d_\psi(z, z_g) \geq 1.25 u_{\max(h, 5)}$, that is, when the TDR places the goal beyond budget h . No label reads the simulator state, so training and inference are both pixel-only; on Push-T, the TDR-derived τ agrees with the success predicate on 97.7% of draws. For $h \geq \Delta$ at unsuccessful states, we sample M blocks around the logged action and use

$$q_h = \max_{m \leq M} \text{clip}_{[0, 1]} \{ \bar{V}(\hat{z}'_m, z_g, h - \Delta) - \kappa \bar{\sigma}(\hat{z}'_m, z_g, h - \Delta) \}, \quad \kappa = 1. \quad (5)$$

A hinge loss penalizes $[V(z, z_g, h) - V(z, z_g, h + \Delta)]_+$. Here $\bar{V}, \bar{\sigma}$ are the EMA ensemble mean and standard deviation. The TDR-derived labels and the ensemble penalty are heuristics: a TDR ball is only a proxy for the goal condition, a TDR distance threshold is not a physical reachability bound, and ensemble spread is not a calibrated confidence interval. We therefore treat the critic as an approximate reachability score and account for its error relative to a specified control process in section 4.

Let B be the budget and t the executed steps. For a full plan with $B - t \geq L$, set $h_{\text{rem}} = \min(50, B - t - L)$ and $n = \lfloor h_{\text{rem}} / \Delta \rfloor$. We sum over all grid points, including both endpoints:

$$\text{ET}(\hat{z}_K, z_g) = \Delta \sum_{j=0}^n [1 - V(\hat{z}_K, z_g, j\Delta)], \quad C_n = (n + 1)\Delta. \quad (6)$$

The cap is C_n , one grid interval beyond the largest queried deadline. When no time remains, the score still penalizes predicted failure at the endpoint. The score equals a truncated expected *grid* hitting time only when all queried probabilities describe the same continuation policy. Otherwise, we use it as a sum of reachability estimates. Updating the budget prevents queries beyond the available time; it does not guarantee success by the deadline.

At each CEM iteration, we normalize each cost over the candidate population using $\tilde{f} = (f - \text{median } f) / s_f$, where $s_f = \max(\text{IQR } f, \nu_f)$. The fixed positive floor ν_f prevents division by zero. The objective is

$$c = \tilde{b} + \beta \tilde{\text{ET}}, \quad \beta = 1. \quad (7)$$

Although β is fixed, the effective relative weight in unnormalized units is $\beta s_b / s_{\text{ET}}$. This ratio is central to the progress bound. The TDR and critic train for 50k and 60k updates; sections B and G.1 give the remaining settings and planning pseudocode.

4 THEORY: FROM THE PLANNING COST TO GOAL PROGRESS

We analyze the normalized graph and critic cost with the trained networks and graph held fixed. Our main result bounds graph progress and local convergence without assuming a globally accurate temporal metric. Section A connects each assumption to the algorithm and provides the proofs. Fix a goal g . Define the remaining graph distance Φ by the attachment objective,

$$\Phi(z) = \min_{v \in A(z)} \{ \|\psi(z) - y_v\| + D_g[v] \}, \quad \mathcal{R}_\lambda = \{ z : d_\psi(z, z_g) \leq \lambda \}. \quad (8)$$

For deterministic execution, we analyze CEM’s returned sequence using the predictor’s input history. The controller ends the episode as soon as the agent satisfies the environment’s goal condition. Section A.5 covers stochastic execution.

Theorem 1 (Normalized-cost graph progress). *Suppose $z \notin \mathcal{R}_\lambda$ has a finite attachment, and let y_q be the first λ -crossing target on its selected path. Write $z^+(u), \hat{z}^+(u)$ for the true and predicted endpoints of an L -step sequence. Suppose a feasible comparison sequence u° satisfies $\|\psi(z^+(u^\circ)) - y_q\| \leq \eta$. For both u° and the executed sequence $\hat{u}$, assume $\|\psi(\hat{z}^+(u)) - \psi(z^+(u))\| \leq e_\psi$. Use the final population’s positive scales s_b, s_S and medians for both costs, and suppose $\hat{c}(\hat{u}) \leq \hat{c}(u^\circ) + \epsilon_{\text{opt}}$. If $[\hat{S}(u^\circ) - \hat{S}(\hat{u})]_+ \leq \omega$ for the ET scores, define*

$$\rho = \eta + 2e_\psi + s_b \epsilon_{\text{opt}} + \beta \frac{s_b}{s_S} \omega. \quad (9)$$

If $\rho \leq r_{TD}$ and $\rho < \lambda$, the controller either reaches the goal during the call, enters $\mathcal{R}_\lambda$, or reduces the remaining graph distance:

$$\Phi(z^+(\hat{u})) \leq \Phi(z) - (\lambda - \rho). \quad (10)$$

If these bounds hold at every graph call before stopping, success or entry into $\mathcal{R}_\lambda$ occurs within $N_G = \lceil \Phi(z_0)/(\lambda - \rho) \rceil$ calls.

An endpoint within r_{TD} of a regular target node can attach there for the next call, reducing the remaining graph distance by at least $\lambda - \rho$. If the target is the goal node, the endpoint enters the local region instead. The bound includes the normalization ratio in equation (9); $\omega = C_n$ always bounds the critic difference but may be too large to prove progress. Median calibration alone does not ensure that the comparison sequence exists. Theorem 17 also shows when the critic rejects poor plans that subgoal distance alone would favor.

Theorem 2 (Local convergence after the switch). *Assume the switched controller stays in $\mathcal{R}_\lambda$ until success and enters with $e(z) = \|z - z_g\| \leq R$. Assume $e(z) \leq r_{\text{goal}}$ implies the environment’s goal condition. At each local call, suppose a feasible comparison sequence reduces true latent error to at most $ae(z)$, where $0 < a < 1$. Suppose endpoint prediction error is at most e_z for the comparison and executed sequences, and the normalized solver and critic bounds of Theorem 1 hold with local constants. Put*

$$d_\ell = 2e_z + \sqrt{s_b \epsilon_{\text{opt}} + \beta(s_b/s_S)\omega_\ell}, \quad e_\infty = \frac{d_\ell}{1-a}. \quad (11)$$

Until success, $e(z_m) \leq a^m R + (1 - a^m)e_\infty$. If $e_\infty < r_{\text{goal}}$, success follows within

$$N_\ell = \max \left\{ 0, \left\lceil \frac{\log(R/(r_{\text{goal}} - e_\infty))}{\log(1/a)} \right\rceil \right\}$$

local calls, with $N_\ell = 0$ if $R = 0$. Together with Theorem 1, $L(N_G + N_\ell)$ is a sufficient execution budget, provided the bounds hold for every remaining-budget value encountered.

These assumptions require the local controller to stay in the switching region and reduce latent error enough to satisfy the task’s goal condition. They play the same role as local control and terminal-region conditions in MPC (Mayne et al., 2000). The reported success rates alone do not establish them.

Theorem 3 (Finite-horizon critic error propagation). *Use a Markov state sufficient to specify block-transition probabilities and the goal condition. After success, keep the process in a successful state. Let V_j^* be the optimal probability of reaching the goal by j blocks: $V_0^* = \mathbf{1}_g$ and $V_j^* = \mathcal{B}V_{j-1}^*$, where $\mathcal{B}$ is the undiscounted reachability Bellman operator. For estimates $\hat{V}_j$, suppose $\|\hat{V}_0 - V_0^*\|_\infty \leq \varepsilon_0$ and $\|\hat{V}_j - \mathcal{B}\hat{V}_{j-1}\|_\infty \leq \xi_j$. Then*

$$\begin{aligned} \|\hat{V}_j - V_j^*\|_\infty &\leq \varepsilon_0 + \sum_{i=1}^j \xi_i, \\ \left| \Delta \sum_{j=0}^n (1 - \hat{V}_j) - \Delta \sum_{j=0}^n (1 - V_j^*) \right| &\leq \Delta \left[(n+1)\varepsilon_0 + \sum_{i=1}^n (n-i+1)\xi_i \right]. \end{aligned} \quad (12)$$

Exact synchronous backups with the boundary held fixed recover horizons through n after n sweeps, from any bounded initialization of the other horizons.

We apply standard reachability dynamic programming (Abate et al., 2008) to this critic. Its residuals include model, action-search, label, and fitting errors; the training loss alone does not bound them. For a common continuation policy π , the sum of survival probabilities is exactly $\mathbb{E}_\pi \min(T^\pi, C_n)$ for grid hitting time T^π . For horizon-wise optimal values it is only a lower bound on $\inf_\pi \mathbb{E}_\pi \min(T^\pi, C_n)$ (Theorem 13). Thus ET need not describe any single policy or guarantee success by the deadline. Section A.9 proves the separate tabular expectile convergence result and explains why it does not imply neural TDR convergence.

Why improvement over terminal L2 is possible. Theorem 23 constructs a problem with an exact model and latents on a circle. A K -block L2 planner stays at the start forever, while the graph objective with an exact critic reaches the goal in two calls, including $K = 5$. Reaching the goal requires a temporary increase in L2 distance, which the short-horizon L2 planner avoids. This example explains one advantage; incorrect edges, poor action coverage, or a conflicting critic can reverse it.

5 EXPERIMENTS

5.1 SETUP

Environments and protocols. We use the public LeWM datasets and checkpoints: 18,685 Push-T expert episodes (2.34M frames), and 10,000 episodes of 201 frames each for Reacher and OGBench Cube (Maes et al., 2026; Park et al., 2025). Push-T success requires agent and block position errors below 20 px and block angle error below $\pi/9$. Reacher uses raw joint-angle differences below 0.05 rad for both joints, without angle wrapping. Cube requires cube-position error below 0.04 m, without an orientation or end-effector condition. The Reacher convention distinguishes unwrapped joint coordinates and therefore need not be identifiable from a single image.

Same-episode goals are $H \in \{25, 50, 100\}$ environment steps after the start, with budget $B = 2H$. Cross-episode tasks draw frames from different episodes, reject pairs already successful at the start, and impose no additional distance or reachability filter. Their budgets are 250 steps on Push-T and 200 on Reacher and Cube. Each protocol uses the same 50 evaluation tasks across methods and five seeds. For our planner, seeds vary TDR, graph, critic, and CEM randomness; L2 seeds vary CEM alone. Standard deviations measure variation across seeds on these fixed tasks. They do not measure uncertainty across new tasks or establish generalization to unseen goals.

Baselines. L2 uses LeWM’s terminal cost. TDR uses $d_\psi(\hat{z}_K, z_g)$ directly. The graph cost-to-go (CTG) baseline uses a terminal attachment cost plus D_g without committing to a subgoal. Random samples environment actions uniformly. We train GCIQL on frozen LeWM latents with the released configuration, ten epochs per phase, and one training seed on Push-T. All planning comparisons use the same main CEM configuration. Panel (B) of table 1 gives published results for context. Those evaluations use different tasks from our matched comparison (section I).

5.2 MAIN RESULTS AND COMPONENT EFFECTS

Table 1 shows gains over LeWM at offsets 50 and 100 and on cross-episode goals in every environment. Push-T gains are 29.2, 33.2, and 28.0 percentage points; Reacher gains are 7.2, 14.8, and 36.4; Cube gains are 15.2, 14.8, and 11.2. At offset 25, the method loses 2.8 points on Push-T and 1.2 on Reacher, and gains 6.0 on Cube. GCIQL reaches 42.0% at offset 25 and 2.0% cross-episode on Push-T; this policy configuration has lower success than the planning methods at both settings.

Push-T ablations. Table 2 isolates the components on the same tasks. TDR alone improves offsets 50 and 100 but leaves cross-episode success near L2. Adding graph subgoals and the final switch reaches 62.4/30.8/17.2% at offsets 50/100/cross-episode. The ET critic on L2 alone reaches 58.4/38.4/31.6%; combining it with the graph and switch reaches 70.0/46.4/42.4%. Thus the graph and critic are complementary on this environment. Switching to L2 locally recovers much of the short-range loss and adds 7.2 points cross-episode when the critic is present.

The $-\log V$ variant has the largest offset-100 mean (47.2%), slightly above ET (46.4%), but loses 10.8 points to ET at offset 25 and 8.0 cross-episode. Integrating survival over several horizons can distinguish candidates that have similar probability at one deadline (Theorem 12). This explanation requires informative horizon predictions; a saturated or miscalibrated critic need not help. The weight sweep in table 6 also shows that $\beta = 1$ is a shared choice rather than an optimal value for every protocol.

Environment dependence. The Cube ablation in table 7 shows a different pattern: TDR alone reaches 36.8% cross-episode, exceeding the full method’s 27.2% in that sweep. Removing the local switch gives the highest offset-100 mean (56.8%). Therefore the full method’s improvement over

Table 1: Success (%). (A) Mean $\pm$ seed standard deviation on 50 fixed tasks and five seeds. Bold marks the best mean. Underlining marks overlap with the best method’s mean $\pm$ standard deviation; it does not indicate statistical significance. [†]GCIQL uses one training seed and a deterministic policy. (B) Maes et al. (2026)’s published results and our offset-25 means use different tasks, so this panel provides context rather than a matched comparison. “–”: not reported.

(A) Our results					
Env	Method	same-ep 25	same-ep 50	same-ep 100	cross-ep
Push-T	Random	4.0 $\pm$ 1.3	0.4 $\pm$ 0.8	0.0 $\pm$ 0.0	0.4 $\pm$ 0.8
	GCIQL [†]	42.0	26.0	22.0	2.0
	L2 (LeWM)	91.2 $\pm$ 1.0	40.8 $\pm$ 3.7	13.2 $\pm$ 3.2	14.4 $\pm$ 3.9
	Ours	<u>88.4</u> $\pm$ 2.3	70.0 $\pm$ 2.2	46.4 $\pm$ 3.2	42.4 $\pm$ 3.2
Reacher	Random	8.0 $\pm$ 3.8	8.4 $\pm$ 0.8	9.2 $\pm$ 3.2	2.8 $\pm$ 1.0
	L2 (LeWM)	77.2 $\pm$ 4.8	90.8 $\pm$ 1.0	82.8 $\pm$ 1.0	29.2 $\pm$ 2.4
	Ours	<u>76.0</u> $\pm$ 4.6	98.0 $\pm$ 1.3	97.6 $\pm$ 0.8	65.6 $\pm$ 2.0
Cube	Random	21.2 $\pm$ 3.0	12.8 $\pm$ 2.0	4.8 $\pm$ 4.1	3.6 $\pm$ 2.3
	L2 (LeWM)	<u>56.0</u> $\pm$ 3.8	34.4 $\pm$ 3.2	35.6 $\pm$ 1.5	16.4 $\pm$ 1.5
	Ours	62.0 $\pm$ 4.7	49.6 $\pm$ 4.6	50.4 $\pm$ 3.2	27.6 $\pm$ 6.1

(B) Reported by Maes et al. (2026) (same-ep 25 only)							
Env	LeWM	PLDM	DINO-WM	GCBC	GCIQL	GCIVL	Ours
Push-T	96	78	74	75	20	33	88.4
Reacher	86	78	79	–	–	–	76.0
Cube	74	65	86	84	64	56	62.0

Table 2: Push-T ablation (success %, same tasks and seeds as table 1). Rows (a)–(c) add TDR, graph subgoals, and the local switch in turn; (d) adds the critic without a graph; (e)–(g) combine both; (h) scores the subgoal with latent L2. Bold and underlining follow table 1. ET sums the survival estimates in equation (6). Table 6 gives the full ablation.

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
L2 (LeWM)	91.2 $\pm$ 1.0	40.8 $\pm$ 3.7	13.2 $\pm$ 3.2	14.4 $\pm$ 3.9
(a) TDR terminal cost (no graph)	83.2 $\pm$ 4.1	48.4 $\pm$ 1.5	22.4 $\pm$ 5.3	14.0 $\pm$ 1.3
(b) graph subgoal	80.8 $\pm$ 2.7	51.6 $\pm$ 2.3	22.4 $\pm$ 5.0	15.2 $\pm$ 3.0
(c) (b) + final switch	<u>88.0</u> $\pm$ 2.5	62.4 $\pm$ 2.7	30.8 $\pm$ 6.5	17.2 $\pm$ 2.7
(d) L2 + ET critic (no graph)	<u>89.2</u> $\pm$ 3.0	58.4 $\pm$ 5.1	<u>38.4</u> $\pm$ 7.1	31.6 $\pm$ 2.3
(e) (b) + ET critic (no switch)	86.0 $\pm$ 3.4	<u>64.8</u> $\pm$ 3.0	<u>43.6</u> $\pm$ 5.0	35.2 $\pm$ 2.0
(f) (c) + $-\log V$ critic	77.6 $\pm$ 1.5	<u>69.2</u> $\pm$ 2.4	47.2 $\pm$ 5.9	34.4 $\pm$ 7.3
(g) (c) + ET critic (ours)	<u>88.4</u> $\pm$ 2.3	70.0 $\pm$ 2.2	<u>46.4</u> $\pm$ 3.2	42.4 $\pm$ 3.2
(h) (g), subgoal scored by L2	86.0 $\pm$ 2.2	<u>63.6</u> $\pm$ 4.8	<u>45.2</u> $\pm$ 3.5	<u>36.4</u> $\pm$ 2.9

LeWM on Cube does not establish that graph routing or the critic improves on TDR. The theorem allows this outcome because it requires accurate local execution and limits how much the critic can oppose target tracking.

5.3 DIAGNOSTICS AND COMPUTATIONAL COST

The Push-T diagnostics distinguish model error from objective ranking. Under logged actions, median predictor error is 2.41 at 25 steps and 7.80 at 100 steps; true start–goal distances are 17.06 and 19.60, respectively (table 9 and figure 3). Replanning every 25 steps limits open-loop error, but does not remove it (table 10). Distances approaching the random-pair scale by 50 steps, suggests that L2 carries less temporal information at longer horizons.

Within the proposed method’s first-iteration CEM elites, the reported mean rank correlations with a simulator recovery-time diagnostic are 0.327/0.203/0.191 at offsets 25/50/100, versus 0.396/0.086/–0.064 for L2 (table 11). This shows that at long horizons L2 cost is a poor ranker,

Table 3: Push-T success (%). **(a)** Increasing position/angle tolerances. **(b)** Cumulative success through each 50-step budget window; parentheses give the fraction of previously unsolved tasks solved in that window (%). Dashes denote time beyond the same-episode-100 budget.

(a) Success threshold			(b) Success by budget window				
threshold	Ours	L2	window (steps)	same-ep 100		cross-ep	
				Ours	L2	Ours	L2
<i>same-ep 100</i>							
20 px/20°	46.4	13.2	0–50	8.8 (8.8)	8.8 (8.8)	14.8 (14.8)	6.0 (6.0)
25 px/25°	52.8	14.8	50–100	23.2 (15.8)	12.0 (3.5)	30.4 (18.3)	12.0 (6.4)
30 px/30°	58.0	16.0	100–150	36.4 (17.2)	12.4 (0.5)	36.4 (8.6)	13.2 (1.4)
			150–200	46.4 (15.7)	13.2 (0.9)	38.8 (3.8)	14.0 (0.9)
			200–250	–	–	42.4 (5.9)	14.4 (0.5)
<i>cross-ep</i>							
20 px/20°	42.4	14.4					
25 px/25°	51.2	19.2					
30 px/30°	57.6	22.8					

Table 4: Adding demonstration initialization to the proposed method: success (%), mean $\pm$ standard deviation on the same 50 tasks and five seeds. Bold marks the larger mean. The main results in table 1 use the default initialization.

Env	Method	same-ep 25	same-ep 50	same-ep 100	cross-ep
Push-T	Ours	88.4 $\pm$ 2.3	70.0 $\pm$ 2.2	46.4 $\pm$ 3.2	42.4 $\pm$ 3.2
	Ours + retrieval	86.8 $\pm$ 2.0	66.8 $\pm$ 5.2	44.8 $\pm$ 4.7	42.4 $\pm$ 6.2
Reacher	Ours	76.0 $\pm$ 4.6	98.0 $\pm$ 1.3	97.6 $\pm$ 0.8	65.6 $\pm$ 2.0
	Ours + retrieval	62.0 $\pm$ 6.4	88.4 $\pm$ 2.7	94.0 $\pm$ 1.3	63.6 $\pm$ 2.3
Cube	Ours	62.0 $\pm$ 4.7	49.6 $\pm$ 4.6	50.4 $\pm$ 3.2	27.6 $\pm$ 6.1
	Ours + retrieval	86.4 $\pm$ 4.1	59.2 $\pm$ 2.4	78.8 $\pm$ 3.0	48.0 $\pm$ 2.8

but outperforms our approach at short horizons, justifying the local switch. Figure 2 shows that straight-line latent midpoints lie mostly inside the observed shell (section F.2). The CEM costs are optimized along this line for every rollout. At offset 100, the arithmetic midpoints have mean radius 10.03, well under the radius of observed latents. Thus a straight line between latent endpoints differs from the path of an observed trajectory. Theorem 23 supplies a separate, explicit dynamical example of L2 failure.

Table 3 shows that our cumulative Push-T success continues to rise over later replans. On cross-episode tasks, success grows from 30.4% at step 100 to 42.4% at step 250, versus 12.0 to 14.4% for L2. Our method is able to maintain sustained increase in success rate over increasing budgets, unlike L2 that stagnates after 50-100 steps. Loosening success tolerances increases both methods’ scores and increases our advantage. This indicates that our method fails with a lesser margin than L2. Demonstration initialization is a useful additional option on Cube: it raises cross-episode success from 27.6 to 48.0%, but helps neither Push-T nor Reacher (table 4 and section E). This variant initializes CEM’s mean with a 25-step logged action sequence whose start is close to the current state and whose endpoint is close to the graph target in TDR space. It retains the cost, candidate count, and execution cadence. The differences across environments show that planning depends on both the sampled actions and the cost used to rank them.

On one GTX 1080 Ti, preparation takes 190.8 GPU-minutes including shared encoding; planning takes 1.80 seconds per decision versus 1.14 for L2, an increase of 58% (table 5). We run Dijkstra’s algorithm once per goal. The deployed TDR and critic add 0.64M and 4.13M reported parameters (table 12).

Theorem 1 explains how repeated calls can make progress when each target is executable and model, search, and critic errors stay bounded along the route. Aggregate success rates do not establish those conditions at every call.

Table 5: Push-T compute on one GTX 1080 Ti. We share encoded frames across seeds and repeat the other preparation stages for each seed. Planning takes $1.58 \times$ L2’s time per decision.

<i>Preparation (GPU-minutes)</i>	
Encode (shared across seeds)	49.6
TDR training (50k steps)	16.6
Graph construction	1.6
Critic training (60k steps)	123.0
Total (incl. shared encode)	190.8
<i>Per-decision planning time (s)</i>	
L2	1.14
Ours (graph subgoal + critic)	1.80
Ratio vs. L2	$1.58 \times$

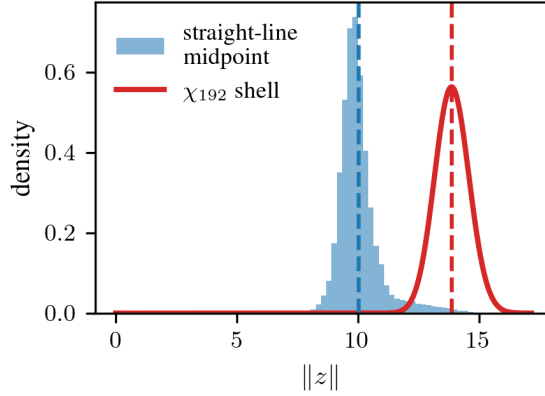


Figure 2: Radii of straight-line latent midpoints for Push-T frame pairs 100 steps apart (blue), with the fitted χ_{192} shell (red). Most midpoints lie inside the shell. We construct these points by interpolation, without running CEM.

6 LIMITATIONS

The critic’s labels come from TDR distances, so their accuracy depends on the TDR: a TDR ball can disagree with the success predicate, and heuristic negatives can mislabel reachable pairs. Label errors, off-distribution model rollouts, finite action proposals, and representation aliasing can violate the theoretical error conditions. The graph links nearby centers and connects the goal using an expanded radius, but these distances do not ensure feasible transitions. The convergence result assumes local control and that sufficiently small latent error implies task success. It does not certify the released checkpoints or prove global convergence of neural training or CEM.

Experiments cover one world-model family and fixed task sets. They do not include matched runs of SAGE, Hi-LeWM, or ALPS. SAGE reports strong results at offset 150 (Cheng et al., 2026), a setting outside our main comparison. We claim neither overall state of the art nor a general advantage over learned subgoal generators. The Cube ablations and short-range losses show where the combined objective fails to help.

7 CONCLUSION

Graph-guided terminal costs improve long-horizon and cross-episode planning with a frozen world model on the evaluated tasks. We bound progress toward graph targets and convergence near the goal using execution, model, search, and critic errors. These bounds explain how the method can improve on terminal L2 when suitable action sequences exist and the controller can find them. The analysis includes the population normalization scales and the critic in both phases. It therefore accounts for the critic’s effect even after the controller switches to local L2, when it must reach the goal accurately.

AI USE STATEMENT

We used generative AI for literature search, mathematical derivation, proof checking, manuscript revision, and document-validation scripts. AI assistance did not produce new benchmark runs; the empirical results are the experimental measurements reported for this study.

ETHICS STATEMENT

The study concerns simulated control benchmarks using public datasets and checkpoints. It involves no human-subject experiment or sensitive personal data.

REPRODUCIBILITY STATEMENT

Sections B and G.1 specify the planning procedure and reported settings; section H gives the rendering check. Section A provides full proofs and states their assumptions. Mathematical validation examples are separate from benchmark measurements. Fixed task sets define the evaluation scope; mathematical checks do not reproduce benchmark runs or establish empirical convergence.

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A DETAILED THEORY AND PROOFS

A.1 SCOPE, STATE, AND TIME CONVENTIONS

We hold the trained encoder, predictor, TDR, graph, and critic fixed throughout the analysis. We examine whether the controller reduces the remaining graph distance, whether this progress brings it to a region where local control reaches the goal, and when the critic’s score measures an expected time to reach the goal. Graph connectivity alone does not establish these properties. All error tolerances and optimization gaps below are nonnegative; the graph radius, lookahead, and normalization floors are positive.

Let x denote a physical state, or a history state sufficient to make the dynamics Markov. Write $z(x)$ for its current encoded observation and $\psi(x) = \psi(z(x))$ when no confusion results. The LeWM predictor uses its input history as well as its current latent. In the deterministic results, $x^+(u)$ is the state after the entire sequence u of $L = K\Delta$ primitive actions, and $z^+(u) = z(x^+(u))$. Write $\hat{z}^+(u)$ for the predicted endpoint. We bound prediction error for the observed input history. The graph-descent theorem therefore does not require a single image latent to be Markov. The Bellman results require a state that contains enough information to be Markov. If the implemented critic receives only $z(x)$, its error must include information lost by that input restriction.

Fix a goal observation z_g . Let $\mathcal{G}_g$ be the measurable set of environment states that satisfy the task’s goal condition. Raw latent distance to z_g need not identify this set. For example, a goal may constrain only an object’s position while images also contain an unconstrained robot configuration. Conversely, two identical rendered arm poses can have different unwrapped joint coordinates. The local theorem explicitly assumes that a small enough latent error implies task success. Matching goal images alone does not ensure that every environment’s goal condition holds. When the agent satisfies the goal condition, the controller ends the episode and skips any remaining actions in the current plan. We need endpoint bounds only for calls that have not already reached the goal. If the agent reaches the goal during a call, it has already satisfied the theorem’s conclusion, even if its latent lies outside the TDR switching region. The graph phase therefore ends when the agent first reaches the goal or enters $\mathcal{R}_\lambda$.

Definition 4 (Block hitting time). Under continuation policy π , let τ^π be the first environment step at which the agent enters $\mathcal{G}_g$, with value ∞ if it never enters. Define $T^\pi = \Delta \lceil \tau^\pi / \Delta \rceil$, including $T^\pi = 0$ for an initially successful state. The block process records whether success occurred at any time within a block and makes that event absorbing.

We count the first time the agent satisfies the goal condition as success; the physical object need not remain at the goal afterward. A block transition that inspects only its final image may miss an earlier success. Such an omission is a modeling error when the intended labels record first hits. The state in the Bellman analysis includes the hit flag when it is needed.

All empirical budgets are in primitive environment steps. The critic grid has spacing $\Delta = 5$, the MPC horizon is $K = 5$ blocks, and the controller executes up to $L = 25$ steps per call, stopping early if it reaches the goal. The main protocols have budgets divisible by L . If the remaining budget is smaller than L , the controller must shorten the plan and update the clock by the number of executed steps, or stop. The full- L formulas apply only when at least L steps remain. We learn the discounted TDR from consecutive-frame transitions; it has its own calibrated units. Neither r_{TD} nor λ is itself a count of environment steps.

Relationship to earlier guarantees. The representation theorem in HILP links embedding and directional-action error to optimal goal-reaching under its stated deterministic assumptions (Park et al., 2024). Our controller instead tracks a node chosen by graph search and executes a whole action sequence. The proof below works with the finite graph’s actual path length, without assuming a globally accurate embedding of optimal temporal distance. PcLast and ALPS use learned representations for hierarchical model-based planning (Koul et al., 2024; Shehmar et al., 2026); the spectral commute-time identity in ALPS concerns representation geometry, whereas we bound progress under the controller’s combined cost. The terminal proof uses the local contraction structure familiar from MPC (Mayne et al., 2000). The reachability recursion is a finite-horizon dynamic program (Abate et al., 2008), which uses different assumptions from infinite-horizon stochastic shortest-path convergence (Bertsekas & Tsitsiklis, 1991).

Correspondence between the algorithm and the assumptions. The potential in equation (8) is the start-attachment objective in equation (3), using the stored TDR centers and the same fixed goal graph. The path length includes the start attachment, as the algorithm’s lookahead rule requires. The comparison sequence contains K action blocks. We compare the frozen predictor’s endpoint with the endpoint after executing all L steps. This assumption requires a suitable action sequence, without requiring a learned low-level policy. We score CEM’s returned mean and the comparison sequence using the final population’s normalization constants. We do not recompute the IQR after adding either sequence. Earlier CEM iterations may use different constants; the proof compares both sequences under the same final objective.

The critic remains in the cost after the local switch, so differences in its scores contribute to both equations (9) and (11). The Bellman reference allows feedback between Δ -step blocks, whereas the controller commits to K such blocks; an accurate reference critic is therefore not automatically the value of this controller. The common-policy survival interpretation requires a separately specified continuation policy. Finally, the tabular exptile theorem analyzes exact value-table updates. TDR training instead uses twin networks, lagged targets, and Euclidean distances. The main progress theorem treats the trained representation as fixed and does not require its optimizer to converge.

If a variant scores the mapped medoid $\tilde{y}_q = \psi(z_q)$ instead of the graph center y_q , let $\delta_q = \|\tilde{y}_q - y_q\|$. If the comparison sequence ends within η of $\tilde{y}_q$, the same argument bounds the executed endpoint’s distance to the graph center by $\rho + \delta_q$; use that quantity in the radius and descent conditions. If η instead measures the comparison sequence’s endpoint distance to y_q , use $\rho + 2\delta_q$. Both statements follow by one triangle inequality at each changed target comparison. Replacing centers by medoids therefore changes the error bound; the formulas must account for the distance between the two targets.

A.2 RESTRICTED ATTACHMENT AND PATH PROGRESS

Let G_g be a finite graph augmented by the goal node, with coordinate $y_g = \psi(z_g)$. Its edges have nonnegative weights equal to Euclidean distances between their coordinates. $D_g[v]$ is finite only for nodes connected to the goal. Choose shortest paths without repeated nodes; such paths exist even with zero-weight edges. Define

$$A(z) = \{v \in V_G : \|\psi(z) - y_v\| \leq r_{\text{TD}}, D_g[v] < \infty\}.$$

The goal is not needed as an extra member of $A(z)$: the explicit local switch handles states near it. If $A(z)$ is empty, $\Phi(z) = +\infty$, and no finite graph-progress bound applies. The following results assume that at least one attachment has a finite route to the goal.

Lemma 5 (Path prefix and attachment). *Let v^* attain equation (8), and let q be a node on a selected shortest path from that attachment to the goal. Let p_q be the cumulative path length from $\psi(z)$ to y_q , including the initial attachment segment. Then*

$$\Phi(z) = p_q + D_g[q], \quad d_\psi(z, z_g) \leq \Phi(z). \quad (13)$$

Here $D_g[g] = 0$. If q is a regular graph node and $\|\psi(z') - y_q\| \leq r_{\text{TD}}$, then

$$\Phi(z') \leq \Phi(z) - p_q + \|\psi(z') - y_q\|. \quad (14)$$

Proof. Every suffix of the selected shortest path is a shortest path from its initial node to the goal. Otherwise replacing that suffix with a shorter one would shorten the selected path. Its total length, including the attachment cost, is $\Phi(z)$, which proves the first identity. Applying the Euclidean triangle inequality successively to the coordinates along the same path proves the second inequality. If the displayed distance from z' to q is at most r_{TD} , the node q is a feasible member of $A(z')$: it has finite $D_g[q]$ because it lies on the route. Evaluating the minimum defining $\Phi(z')$ at q proves equation (14). $\square$

The proof does not assume that D_g has a Lipschitz extension or that Φ is continuous. The radius restriction can make Φ jump when a node enters or leaves the attachment set. We therefore require the next endpoint to lie within r_{TD} of the selected node. A qualitative statement that the endpoint is nearby would not suffice.

Lemma 6 (First-crossing lookahead). *Suppose $z \notin \mathcal{R}_\lambda$. The selected path has length greater than λ and therefore contains a first λ -crossing target. If r_e is the largest segment on the augmented path, including the initial attachment, then*

$$\lambda \leq p_q < \lambda + r_e.$$

One may take $r_e \leq \max(r_{TD}, r_g)$ for the stated graph construction.

Proof. By Theorem 5, $\Phi(z) \geq d_\psi(z, z_g) > \lambda$. The finite cumulative sequence starts at zero and ends at $\Phi(z)$. Just before its first crossing it is strictly below λ ; the crossing adds one segment of length at most r_e . Regular edges and the initial attachment are at most r_{TD} , and goal-attachment edges are at most r_g . $\square$

The selected target can therefore lie beyond the calibrated lookahead distance. In particular, expanding the goal-attachment radius can create a much longer last edge. The comparison sequence in Theorem 1 must reach near this actual target, including any extra distance from crossing the lookahead threshold. Likewise, resetting cluster centers to means after sequential admission does not preserve an admission-radius guarantee for all members. The proof needs the actual center distances and endpoint error, not an unproved bound on cluster radii.

Lemma 7 (Graph descent from endpoint tracking). *Suppose a first-crossing target is tracked with error $\|\psi(z') - y_q\| \leq \rho$, where $\rho \leq r_{TD}$ and $\rho < \lambda$. Then either $z' \in \mathcal{R}_\lambda$ or $\Phi(z') \leq \Phi(z) - \lambda + \rho$.*

Proof. If q is regular and z' is outside $\mathcal{R}_\lambda$, apply Theorem 5 and $p_q \geq \lambda$. If $q = g$, the tracking error is $d_\psi(z', z_g) \leq \rho < \lambda$, so $z' \in \mathcal{R}_\lambda$. These cases exhaust the possibilities. $\square$

Why geometric edges require an execution assumption. An undirected radius graph can connect two latent clusters even if the physical dynamics permit travel in only one direction, or no travel at all. The temporal-efficiency filter concerns observed directions within episodes and does not prove reversibility of arbitrary inter-cluster edges. A finite D_g therefore does not ensure that the agent can follow the route. The comparison-sequence assumption requires an executable plan to the chosen target. It allows other graph edges to be infeasible, but requires more than proximity between the current state and target coordinates.

A.3 THE NORMALIZED CEM COST AND EXECUTION ERROR

Fix one planning call, target, horizon grid, and set of normalization constants. Write

$$\hat{b}(u) = \|\psi(\hat{z}^+(u)) - y_q\|, \quad \hat{S}(u) = \Delta \sum_{j=0}^n [1 - V(\hat{z}^+(u), z_g, j\Delta)].$$

Since the critic outputs lie in $[0, 1]$, $0 \leq \hat{S}(u) \leq C_n$ even if they are not monotone in the horizon. Denote the two population medians by m_b, m_S .

Lemma 8 (Exact scale-ratio comparison). *If*

$$\frac{\hat{b}(\hat{u}) - m_b}{s_b} + \beta \frac{\hat{S}(\hat{u}) - m_S}{s_S} \leq \frac{\hat{b}(u^\circ) - m_b}{s_b} + \beta \frac{\hat{S}(u^\circ) - m_S}{s_S} + \epsilon_{\text{opt}},$$

with $s_b, s_S > 0$ and $\beta \geq 0$, then

$$\hat{b}(\hat{u}) \leq \hat{b}(u^\circ) + s_b \epsilon_{\text{opt}} + \beta \frac{s_b}{s_S} [\hat{S}(u^\circ) - \hat{S}(\hat{u})]_+. \quad (15)$$

Proof. Cancel the medians, move the critic difference to the right, and multiply by s_b . Replacing a negative critic difference by zero can only enlarge the right side. $\square$

Normalization still leaves a tradeoff between the two costs. For example, multiplying raw critic scores by a positive constant generally cancels through their IQR when the floor is inactive, but a very small critic IQR can amplify small, unreliable variations. The numerical floors make the objective defined; they do not establish that the resulting scale ratio is appropriate. If two candidates use different population medians or scales, those constants do not cancel and the comparison above no longer applies.

Proof of Theorem 1. If the agent reaches the goal before the call ends, the conclusion already holds. Otherwise, the triangle inequality and the model-error assumption give

$$\widehat{b}(u^\circ) \leq \|\psi(z^+(u^\circ)) - y_q\| + \|\psi(\hat{z}^+(u^\circ)) - \psi(z^+(u^\circ))\| \leq \eta + e_\psi.$$

Apply Theorem 8, then bound the distance between the predicted and true endpoints of the returned sequence:

$$\begin{aligned} \|\psi(z^+(\hat{u})) - y_q\| &\leq \widehat{b}(\hat{u}) + e_\psi \\ &\leq \eta + 2e_\psi + s_b \epsilon_{\text{opt}} + \beta(s_b/s_S)\omega = \rho. \end{aligned}$$

Theorem 7 proves the one-call claim. For the iteration bound, let τ_G be the first call that reaches the goal or has an endpoint in $\mathcal{R}_\lambda$, and let $a_G = \lambda - \rho > 0$. If $\tau_G > N$, all of the first N transitions remain outside $\mathcal{R}_\lambda$, so induction gives $0 \leq \Phi(z_N) \leq \Phi(z_0) - Na_G$. For $N = \lceil \Phi(z_0)/a_G \rceil$, the right side is nonpositive. Equality is also impossible outside $\mathcal{R}_\lambda$, since $\Phi(z_N) \geq d_\psi(z_N, z_g) > \lambda > 0$. Thus $\tau_G \leq N$. If the initial state is already successful or in $\mathcal{R}_\lambda$, take $N_G = 0$. $\square$

What the optimization assumption means for CEM. The comparison need not assume a global minimizer. It can use any feasible sequence u° whose tracking error is controlled, provided the returned sequence has the stated objective gap relative to that sequence. CEM’s final mean is not necessarily a sampled candidate or an elite, and a mean of successful sequences can fail in a non-linear system. Therefore a proof concerning only the best elite would not prove the theorem for the stated implementation. Here ϵ_{opt} compares the sequence the controller actually executes with the feasible comparison sequence, using the final population’s fixed constants. Thirty CEM iterations do not establish that this gap is small (de Boer et al., 2005). One could evaluate the returned mean and retain a candidate with a lower score. The reported runs do not establish that they used this additional check.

Proposition 9 (Action coverage under adaptive sampling). *Let $\mathcal{U}_\eta$ be sequences meeting a prescribed true tracking tolerance. Suppose every conditional CEM proposal, given all previous draws, places probability at least $p > 0$ on $\mathcal{U}_\eta$. Among N draws the probability of seeing no such sequence is at most $(1 - p)^N$.*

Proof. Let F_i be the event that the first i proposals miss $\mathcal{U}_\eta$. Conditional on the past and F_{i-1} , the next failure probability is at most $1 - p$. The tower property yields $\mathbb{P}(F_i) \leq (1 - p)\mathbb{P}(F_{i-1})$. Induction proves the result; independence between iterations is not required. $\square$

This result bounds the chance of sampling a useful sequence under the stated coverage assumption. If CEM concentrates its Gaussian proposals too narrowly, the lower bound p may fail. Even when CEM samples a good sequence, its final mean need not be good. Retrieval changes which actions CEM is likely to sample. It can therefore help independently of the cost, as the Cube results suggest.

Prediction error over one planning call. Suppose true and predicted histories admit a common normed state space with block transitions f and $\hat{f}$, the initial history error is zero, and

$$\|\hat{f}(x, a) - f(x, a)\| \leq \varepsilon_P, \quad \|f(x, a) - f(x', a)\| \leq L_f \|x - x'\|$$

on all compared rollout states and actions. Applying both transitions to the same action sequence and using induction gives

$$e_K \leq \varepsilon_P \sum_{i=0}^{K-1} L_f^i. \tag{16}$$

Indeed $e_{k+1} \leq \varepsilon_P + L_f e_k$ by adding and subtracting the true transition from the predicted state. If the map from history to the endpoint TDR coordinate is L_ψ -Lipschitz, take $e_\psi \leq L_\psi e_K$. The bound is $K\varepsilon_P$ for $L_f = 1$, is bounded uniformly in K for $L_f < 1$, and can grow geometrically for $L_f > 1$. These assumptions suffice to bound endpoint error. A small latent prediction loss alone does not imply them, especially if the encoded history cannot determine the next state. The graph supports progress across many short calls, while this model bound covers only one call. Model-based RL analyses also use short rollouts to limit accumulated error (Janner et al., 2019). Here we do not assume globally stable LeWM dynamics. Each replan starts from new observations, but the endpoint bound must still hold at every state the controller reaches.

A.4 TERMINAL-REGION CONVERGENCE AND THE COMPLETE BUDGET

The switched branch uses squared raw latent distance, while the graph branch uses an unsquared TDR distance. The distinction changes the optimization-error term from linear to square-root dependence. We include the critic in the local-phase proof because the controller still uses it.

Proof of Theorem 2. If a call succeeds before its endpoint, the success conclusion already holds. At a local state of error e , let u° be the contracting comparison sequence. Its true endpoint error is at most ae , and its predicted endpoint error is at most $ae + e_z$. Theorem 8, now applied to the squared base cost, gives

$$\|\hat{z}^+(\hat{u}) - z_g\|^2 \leq (ae + e_z)^2 + s_b \epsilon_{\text{opt}} + \beta(s_b/s_S)\omega_\ell.$$

All terms added on the right are nonnegative. Taking square roots and using $\sqrt{v^2 + w} \leq v + \sqrt{w}$ for $v, w \geq 0$, followed by the endpoint model bound, yields

$$e(z^+(\hat{u})) \leq ae + 2e_z + \sqrt{s_b \epsilon_{\text{opt}} + \beta(s_b/s_S)\omega_\ell} = ae + d_\ell.$$

Invariance ensures that the next call again uses this branch and its assumptions. Iteration gives $e(z_m) \leq a^m e(z_0) + d_\ell \sum_{i=0}^{m-1} a^i \leq a^m R + (1 - a^m)e_\infty$. It suffices that $a^m R \leq r_{\text{goal}} - e_\infty$, which is precisely the stated conservative bound on m . If $R = 0$, entry is already successful. The assumption that latent error within r_{goal} implies task success proves the claim. Adding the graph and local call counts proves the execution-budget bound. Reaching the goal during either phase only shortens the execution. $\square$

Ensuring that the controller stays in the local region. One sufficient special case is that, on a local domain, $\mathcal{R}_\lambda = \{z : e(z) \leq R\}$ and $aR + d_\ell \leq R$. The contraction inequality then leaves the state in $\mathcal{R}_\lambda$. More generally, each local call must end inside the TDR switching region until success. Small raw latent error alone does not ensure this; one must relate raw latent distance to TDR distance. If the implemented controller exits $\mathcal{R}_\lambda$, the local proof stops and the graph phase can resume. Neither theorem alone rules out repeated switching. The assumption that local calls stay in the region is therefore necessary for this proof.

Success tolerance and an unavoidable error floor. With persistent model or optimization error, the bound approaches e_∞ rather than zero. The strict condition $e_\infty < r_{\text{goal}}$ ensures that the error bound eventually enters the success tolerance. If $e_\infty = r_{\text{goal}}$ and R is larger, the upper bound approaches the boundary without reaching it in finite time. If unsuccessful states can lie arbitrarily close to the goal latent, no positive latent radius guarantees task success. This matters when the goal depends on state coordinates that the image does not reveal.

Clock dependence and available time. The ET grid and its IQR change with $B - t - L$. To combine Theorems 1 and 2, choose error constants that cover every call at its remaining budget. Values measured only at the initial budget need not suffice. If $B < L(N_G + N_\ell)$ the theorem does not assert success within the episode, although the controller may succeed faster than the bound. Increasing the number of steps executed before replanning requires a new model-error bound.

A.5 STOCHASTIC EXECUTION AND EXPECTED PROGRESS

We now allow individual calls to violate the deterministic assumptions with bounded probability. The first result bounds success probability within a fixed budget by bounding the chance of any such violation. The second bounds the expected number of graph calls and also requires a limit on how much a failed call can increase the remaining distance.

Corollary 10 (Finite-budget success event). *Let $N = N_G + N_\ell$ bound the number of calls whenever all deterministic assumptions hold. Conditional on earlier calls satisfying their bounds and on their history, suppose call j obeys the appropriate one-call error and comparison bounds with probability at least $1 - p_j$. Include the bound $e \leq R$ on local entry and retention in $\mathcal{R}_\lambda$ for local calls that do not succeed. Take $0 \leq p_j \leq 1$. If $B \geq LN$, success probability is at least $1 - \sum_{j=1}^N p_j$ (or zero when that lower bound is negative).*

Proof. The probability that the first violated call occurs at j is at most p_j , by conditioning on the history up to that call. These first-failure events are disjoint. On their complement the deterministic bound applies for all required calls. Summing their probabilities proves the lower bound. Once the agent reaches the goal, no later call can fail because the episode has ended. The proof does not require independence between calls. $\square$

An ensemble standard deviation alone does not give the p_j values. To apply the corollary, one must justify the conditional execution or model-error probabilities separately; the corollary then combines them.

Proposition 11 (Expected graph-phase duration). *Let τ_G be the first call with success or local-region entry, and set $W_j = \Phi(z_j)$ before τ_G and $W_j = 0$ afterward, for a fixed initial state. Assume these stopped potentials are finite and integrable. Before τ_G , suppose a good call decreases W by at least $a_G > 0$ with conditional probability at least $1 - p$, while any other call increases it by at most M . Here $0 \leq p \leq 1$ and $M \geq 0$. If $\delta = (1 - p)a_G - pM > 0$, then τ_G satisfies*

$$\mathbb{E}[\tau_G] \leq W_0/\delta.$$

Proof. For $j < \tau_G$, the conditional expected increment is at most $-(1 - p)a_G + pM = -\delta$. After stopping the increment is zero. Hence

$$\mathbb{E}W_{(n+1) \wedge \tau_G} \leq \mathbb{E}W_{n \wedge \tau_G} - \delta \mathbb{P}(\tau_G > n).$$

Sum for $n = 0, \dots, m-1$ and use nonnegativity of the left-hand potential: $\delta \sum_{n=0}^{m-1} \mathbb{P}(\tau_G > n) \leq W_0$. Monotone convergence and the integer tail-sum identity give the result. $\square$

For the deterministic progress bound, $a_G = \lambda - \rho \leq \lambda$. Reaching the goal or entering $\mathcal{R}_\lambda$ also gives the required decrease: outside that set $\Phi(z) > \lambda \geq a_G$, so resetting W to zero decreases it by at least a_G . If a failed call can lose all finite graph attachments, no finite M follows. This result therefore does not cover calls that disconnect the agent from the goal graph.

A.6 WHAT THE SURVIVAL SCORE MEASURES

For a specified policy π , write $F_j^\pi(x) = \mathbb{P}_x^\pi(T^\pi \leq j\Delta)$ and $S_n^\pi(x) = \Delta \sum_{j=0}^n (1 - F_j^\pi(x))$. The probabilities are nondecreasing in j because the underlying first-hit events are nested.

Proposition 12 (Survival identity, truncation, and ranking). *For any such policy, including one with a positive probability of never hitting,*

$$S_n^\pi(x) = \mathbb{E}_x^\pi[\min(T^\pi, C_n)], \quad C_n = (n+1)\Delta. \quad (17)$$

If $F_j^a \geq F_j^b$ for every queried j , then $S_n^a \leq S_n^b$. If one inequality is strict, the score inequality is strict. Two states with the same final-deadline probability can nevertheless have different S_n .

Proof. For every $T \in \{0, \Delta, 2\Delta, \dots, \infty\}$, the pointwise identity

$$\min(T, (n+1)\Delta) = \Delta \sum_{j=0}^n \mathbf{1}_{\{T > j\Delta\}}$$

counts the grid intervals before truncation. Because the sum is finite, we can take expectations term by term even when $\mathbb{E}T$ is infinite. This proves the first claim. Subtracting the sums proves the dominance statements. For the last claim, take deterministic times Δ and 2Δ with $n \geq 2$: both final probabilities are one but their scores are Δ and 2Δ . $\square$

The endpoint convention. The grid $j = 0, \dots, n$ includes a term at the final queried deadline $n\Delta$. Truncating the hitting time at that deadline would require summing only $j = 0, \dots, n-1$, giving zero when $n = 0$. Our score retains the last term and therefore has cap C_n . At zero remaining time, it is $\Delta(1 - V_0)$, an endpoint-success penalty. At the maximum query horizon of 50 steps it has cap 55 steps. This extra interval changes rankings whenever V_n differs across candidates; so removing it can change which candidate the planner prefers.

If τ^π is the primitive hitting time, rounding introduces a bounded discrepancy: for any grid-aligned cap C ,

$$0 \leq \mathbb{E} \min(\Delta \lceil \tau^\pi / \Delta \rceil, C) - \mathbb{E} \min(\tau^\pi, C) \leq \Delta.$$

This follows pointwise from $0 \leq \Delta \lceil \tau / \Delta \rceil - \tau < \Delta$ at finite τ , with both capped quantities equal at $\tau = \infty$. Even with exact grid probabilities, the score measures time rounded up to the next block boundary, with the discrepancy bounded above.

Proposition 13 (Horizon-wise optima and policy consistency). *Let Π be a common policy class and $V_j^*(x) = \sup_{\pi \in \Pi} F_j^\pi(x)$. Define*

$$S_n^*(x) = \Delta \sum_{j=0}^n (1 - V_j^*(x)), \quad C_n^*(x) = \inf_{\pi \in \Pi} S_n^\pi(x).$$

Then $S_n^ \leq C_n^*$ and, with $A_\pi = \Delta \sum_{j=0}^n (V_j^* - F_j^\pi)$,*

$$C_n^* - S_n^* = \inf_{\pi \in \Pi} A_\pi \geq 0. \quad (18)$$

Equality holds if a single policy achieves all the horizon optima. More generally, if some $\bar{\pi}$ satisfies $0 \leq V_j^ - F_j^{\bar{\pi}} \leq a_j$, then $0 \leq C_n^* - S_n^* \leq \Delta \sum_{j=0}^n a_j$.*

Proof. For every fixed π , subtract the finite sums to obtain $S_n^\pi = S_n^* + A_\pi$. Each summand in A_π is nonnegative. Taking the infimum proves the identity and lower bound. Substituting a jointly optimal policy, or the approximate policy $\bar{\pi}$, gives the remaining claims. An infimum need not be attained for the identity to hold. $\square$

A strict policy-consistency gap. Take $\Delta = 1$, $n = 2$, and an initial state with two irreversible choices. Choice A reaches the goal at time 1 with probability 0.6 and never reaches it otherwise. Choice B reaches the goal at time 2 with certainty. There are no later choices. Then $V_0^* = 0$, $V_1^* = 0.6$, and $V_2^* = 1$, so $S_2^* = 1.4$. Policy A has truncated expectation $0.6 \cdot 1 + 0.4 \cdot 3 = 1.8$; policy B has expectation 2. Randomizing between them gives a convex combination, so $C_2^* = 1.8 > S_2^*$. The values $j \mapsto V_j^*$ never decrease with the horizon and define a valid cumulative distribution on this grid. Yet no policy in this example has that hitting-time distribution. A critic can therefore be nondecreasing in the horizon without describing any single policy.

Expected truncated time is not deadline success. Even with a common policy interpretation, minimizing the score is not equivalent to maximizing probability of success by the final deadline. With deadline 5 and cap 6, one choice hits at time 1 with probability 0.49 and never hits otherwise; another hits at time 5 with certainty. Their scores are $0.49 + 0.51 \cdot 6 = 3.55$ and 5, respectively. The score prefers the less reliable choice because it assigns enough value to early success. Minimizing this cost therefore does not impose a lower bound on the probability of success by the deadline.

Corollary 14 (Clock sensitivity). *For $n' \geq n$ and any $[0, 1]$ -valued horizon predictions,*

$$0 \leq \widehat{S}_{n'}(x) - \widehat{S}_n(x) \leq \Delta(n' - n).$$

The difference of the score changes for two candidates has absolute value at most $\Delta(n' - n)$.

Proof. The added terms form a sum of $n' - n$ numbers in $[0, \Delta]$. Each candidate's increase lies in the same interval, so their difference lies between its negative and positive endpoints. $\square$

Failing to update the remaining budget can therefore change candidate rankings. If reachability predictions increase with the horizon, querying beyond the available time can also overstate the probability of success by the actual deadline. Changing the horizon can additionally change the score IQR and its effective weight in the combined cost.

A.7 FINITE-HORIZON BELLMAN ANALYSIS OF THE CRITIC

On the stopped Markov block process, define

$$(\mathcal{B}f)(x) = \begin{cases} 1, & x \in \mathcal{G}_g, \\ \sup_{a \in \mathcal{A}(x)} \mathbb{E}[f(X') \mid x, a], & x \notin \mathcal{G}_g. \end{cases} \quad (19)$$

Here a is a Δ -step action block, and X' includes within-block success. Assume measurable policies and kernels for which the finite-horizon recursion is valid; finite state/action spaces suffice. Keep the same action class throughout the comparison. This operator describes optimal reachability using the allowed block actions. It need not equal optimal control with feedback at every primitive step.

Lemma 15 (Nonexpansiveness). *For bounded f, g , $\|\mathcal{B}f - \mathcal{B}g\|_\infty \leq \|f - g\|_\infty$. The corresponding fixed-policy operator has the same property.*

Proof. On the success set both outputs equal one. Elsewhere every conditional expectation differs by at most $\|f - g\|_\infty$. The inequality $|\sup_a u_a - \sup_a v_a| \leq \sup_a |u_a - v_a|$ proves the claim. Taking the expectation under a fixed action distribution gives the policy case. $\square$

This operator does not contract errors by a discount factor. The recursion converges because each horizon depends only on the preceding horizon and we fix the value at horizon zero.

Proof of Theorem 3. Set $e_j = \|\hat{V}_j - V_j^*\|_\infty$. Insert $\mathcal{B}\hat{V}_{j-1}$ and use Theorem 15 to obtain $e_j \leq \xi_j + e_{j-1}$. Induction gives $e_j \leq \varepsilon_0 + \sum_{i=1}^j \xi_i$. The triangle inequality bounds the survival-score difference by $\Delta \sum_{j=0}^n e_j$. Reordering the finite double sum yields

$$\sum_{j=0}^n \sum_{i=1}^j \xi_i = \sum_{i=1}^n (n - i + 1) \xi_i,$$

proving equation (12). For the sweep claim, hold $V_0^{(k)} = V_0^*$ and update $V_j^{(k+1)} = \mathcal{B}V_{j-1}^{(k)}$. After the first sweep V_1 is exact, after the second V_2 is exact, and so on. An induction in j proves exactness through horizon n after n sweeps. $\square$

If $\xi_j \leq \xi$ uniformly, the score-error bound becomes $\Delta[(n+1)\varepsilon_0 + n(n+1)\xi/2]$. We can tighten it by replacing each e_j with $\min(1, \varepsilon_0 + \sum_{i \leq j} \xi_i)$ when both values lie in $[0, 1]$. Probability error therefore grows at most linearly with the horizon under a uniform residual bound; summing those errors across horizons gives a quadratic bound.

Decomposing a residual of the actual bootstrap. Let $\mathcal{B}_M$ be the exact-model Bellman operator restricted to the sampled action blocks, and let $\hat{\mathcal{B}}_M$ use predicted endpoints and the clipped pessimistic ensemble target. For a target function $\bar{V}_{j-1}$, the triangle inequality gives

$$\begin{aligned} \|\hat{V}_j - \mathcal{B}\hat{V}_{j-1}\|_\infty &\leq \underbrace{\|\hat{V}_j - \hat{\mathcal{B}}_M \bar{V}_{j-1}\|_\infty}_{\text{fitting and mixed-label error}} + \underbrace{\|\hat{\mathcal{B}}_M \bar{V}_{j-1} - \mathcal{B}_M \bar{V}_{j-1}\|_\infty}_{\text{model and pessimism error}} \\ &\quad + \underbrace{\|\mathcal{B}_M \bar{V}_{j-1} - \mathcal{B}\bar{V}_{j-1}\|_\infty}_{\text{action-coverage error}} + \underbrace{\|\bar{V}_{j-1} - \hat{V}_{j-1}\|_\infty}_{\text{target lag}}. \end{aligned} \quad (20)$$

The last term follows from nonexpansiveness. Take suprema over a domain that contains every successor of each compared transition. If the initial domain is smaller, extend the error bound to every queried successor. A finite training loss alone does not bound these worst-case errors.

For example, in deterministic dynamics suppose the true and predicted block endpoints are within e_P , the target value is L_V -Lipschitz on the comparison domain, and the ensemble spread is at most $\bar{\sigma}$. Assume both operators enforce the same goal boundary and their endpoint descriptions include the within-block hit flag. The model-and-pessimism term is at most $L_V e_P + \kappa \bar{\sigma}$: each action value changes by at most this amount, clipping to $[0, 1]$ is nonexpansive, and taking a maximum preserves the bound. For a stochastic transition, a deterministic mean prediction generally does not preserve the expectation of a nonlinear value function. One must instead bound the error in the expected next-state value, for example using a total variation bound for $[0, 1]$ -valued functions. Include that error in the same model residual term.

Finite action maximization. If, for a fixed query, a proposal hits an ϵ_A -optimal block with probability at least p_A , M independent draws miss that set with probability $(1 - p_A)^M$. On a hit, the restricted maximum is within ϵ_A of the full maximum for that query. The conditional version of Theorem 9 also applies. Sampling around logged actions does not by itself supply a positive lower bound on coverage of all optimal actions. A uniform Bellman residual must account for this restriction.

Boundary and absorbing-goal errors. The recursion requires $V_0 = \mathbf{1}_g$ and the value one on already-successful states at every horizon. If a target omits either boundary condition, include the resulting error in ϵ_0 or ξ_j . Hindsight at a logged future goal gives a successful behavior trajectory, but observing no hit by a deadline does not prove no other policy can hit. Similarly, an object displacement larger than any displacement observed in 50 steps does not prove physical impossibility in 50 steps. The mixed training procedure therefore approximates reachability, and Theorem 3 accounts for its error through the Bellman residual.

A.8 FROM CRITIC CALIBRATION TO CANDIDATE RANKING

The graph theorem bounds the critic-score difference by ω without requiring accurate probabilities. When the critic does estimate reachability accurately, we can relate this difference to the true endpoint scores. Fix any true endpoint score $S(u)$ of interest: a specified policy’s truncated expectation, or the score formed from the optimal reachability values at each horizon. Suppose on all compared sequences

$$|\hat{S}(u) - S(u)| \leq e_S, \quad |\hat{b}(u) - b(u)| \leq e_b. \quad (21)$$

For the graph branch, $e_b \leq e_\psi$ suffices. If critic error at true endpoints at horizon j is at most ϵ_j and the estimated critic is L_j -Lipschitz between the true and predicted endpoints, then an endpoint prediction bound e_z gives

$$e_S \leq \Delta \sum_{j=0}^n (\epsilon_j + L_j e_z). \quad (22)$$

This follows by adding and subtracting each critic value at its true endpoint and summing absolute differences. Checking critic accuracy only on encoded dataset frames does not control the additional error at predicted endpoints.

Proposition 16 (Candidate regret with fixed population scales). *Fix positive scales and define $J(u) = b(u)/s_b + \beta S(u)/s_S$ and $\hat{J}(u) = \hat{b}(u)/s_b + \beta \hat{S}(u)/s_S$. On a comparison set $\mathcal{U}$, suppose equation (21) holds and $\hat{J}(\hat{u}) \leq \inf_{u \in \mathcal{U}} \hat{J}(u) + \epsilon_{\text{opt}}$, with $\hat{u} \in \mathcal{U}$. Then*

$$J(\hat{u}) \leq \inf_{u \in \mathcal{U}} J(u) + \epsilon_{\text{opt}} + 2 \left(\frac{e_b}{s_b} + \beta \frac{e_S}{s_S} \right). \quad (23)$$

For a comparison sequence $u^\circ \in \mathcal{U}$ that tracks the graph target, use $\omega = [S(u^\circ) - S(\hat{u})]_+ + 2e_S$ in Theorem 1, provided its other hypotheses hold.

Proof. Let $e_J = e_b/s_b + \beta e_S/s_S$. Then $|J(u) - \hat{J}(u)| \leq e_J$ throughout $\mathcal{U}$, so

$$J(\hat{u}) \leq \hat{J}(\hat{u}) + e_J \leq \inf_{\mathcal{U}} \hat{J} + \epsilon_{\text{opt}} + e_J \leq \inf_{\mathcal{U}} J + \epsilon_{\text{opt}} + 2e_J.$$

No minimizer need exist because the inequalities preserve infima. For the last claim, $\hat{S}(u^\circ) - \hat{S}(\hat{u}) \leq S(u^\circ) - S(\hat{u}) + 2e_S$. Take positive parts and use $[a + b]_+ \leq [a]_+ + b$ for $b \geq 0$. $\square$

This result compares costs within a fixed set of action sequences using shared scales. It does not compare against an unrestricted optimal RL policy or imply greater success by the deadline. For our method, it shows that the actual IQR scale ratio multiplies critic error in the graph-progress bound. Even small probability errors can matter when the critic score has a very small normalization scale.

Proposition 17 (When the critic resolves ambiguous subgoals). *Partition a comparison set $\mathcal{U}$ into acceptable and unacceptable plans, and let u° be one acceptable plan. Assume equation (21) and shared positive scales. For every unacceptable u , suppose*

$$b(u) - b(u^\circ) \geq -a_b, \quad S(u) - S(u^\circ) \geq m_S,$$

where $a_b \geq 0$ and $m_S > 0$. If the returned sequence belongs to $\mathcal{U}$, is within ϵ_{opt} of u° in predicted combined cost, and

$$\beta \frac{m_S - 2e_S}{s_S} > \frac{a_b + 2e_b}{s_b} + \epsilon_{\text{opt}}, \quad (24)$$

then the returned plan is acceptable. If every acceptable plan tracks the graph target within $\rho_0 \leq r_{\text{TD}}$, $\rho_0 < \lambda$, the graph-descent conclusion holds with ρ_0 in place of equation (9).

Proof. For an unacceptable plan, subtract the predicted objective at u° . The shared medians cancel, and the error bounds give

$$\hat{J}(u) - \hat{J}(u^\circ) \geq -\frac{a_b + 2e_b}{s_b} + \beta \frac{m_S - 2e_S}{s_S} > \epsilon_{\text{opt}}.$$

This contradicts the stated comparison inequality if the returned plan is unacceptable. The final claim is Theorem 7. $\square$

This condition shows how the critic can help select a useful plan. The base distance may favor an unacceptable plan by up to a_b . If the critic estimates the survival-score gap accurately enough and has enough weight, it reverses that ordering. For a common continuation policy, if an acceptable endpoint has grid hitting time at most $t_0 \leq n\Delta$ almost surely and an unacceptable endpoint cannot hit by $n\Delta$, their score gap is at least $C_n - t_0$. By contrast, if every candidate is unable to hit before truncation, all exact scores are C_n and the critic has no ranking margin. This gives a concrete reason for combining a short-horizon critic with graph direction. Applying these inequalities requires a true score gap and access to acceptable action sequences. Ensemble training alone does not guarantee either condition.

Supervised labels and population risk. Binary cross-entropy is a strictly proper scoring rule (Gneiting & Raftery, 2007). It identifies the conditional probability of the training labels, which need not equal optimal reachability. This distinction can be quantified.

Proposition 18 (Label bias and distribution shift). *Let μ be a distribution of state-goal-horizon queries, let $q(w) = \mathbb{E}[Y | w]$, and let $\hat{V}(w) \in (0, 1)$ be any predictor. The label Y may be binary or a soft target in $[0, 1]$. Let R_{CE} be its population cross-entropy excess over q . For a query distribution ν with $d\nu/d\mu \leq C_\mu$ and a target reachability $F(w)$,*

$$\mathbb{E}_\nu |\hat{V} - F| \leq \sqrt{C_\mu R_{\text{CE}}/2} + \mathbb{E}_\nu |q - F|. \quad (25)$$

Proof. Conditional cross-entropy is linear in Y , so replacing Y by its conditional mean gives excess $\text{kl}(\text{Ber}(q) \parallel \text{Ber}(\hat{V}))$. For fixed $\hat{v} \in (0, 1)$, this divergence as a function of q has value and first derivative zero at $q = \hat{v}$, and second derivative $1/[q(1-q)] \geq 4$ on $(0, 1)$. Twice integrating this curvature inequality, and taking limits at the endpoints, gives $\text{kl}(q \parallel \hat{v}) \geq 2(q - \hat{v})^2$. Therefore $\mathbb{E}_\mu (\hat{V} - q)^2 \leq R_{\text{CE}}/2$. Change measure, use Cauchy-Schwarz, and then the triangle inequality:

$$\mathbb{E}_\nu |\hat{V} - F| \leq \sqrt{\mathbb{E}_\nu (\hat{V} - q)^2} + \mathbb{E}_\nu |q - F| \leq \sqrt{C_\mu R_{\text{CE}}/2} + \mathbb{E}_\nu |q - F|. \quad \square$$

Increasing network capacity or training time does not remove label bias. Hindsight behavior labels, selected cross-episode negatives, and pessimistic model targets can induce different conditional targets q . The result bounds the cross-entropy component for a fixed predictor and label distribution. The training procedure also uses a monotonicity penalty and moving targets, so it does not simply minimize that fixed risk. Moreover, a small average error does not give the uniform bound required by Theorem 16. CEM may select rare states where the critic makes large errors.

Proposition 19 (Finite-query calibration audit). *Fix a trained critic and m query states, selected independently of a validation sample. For each query collect N independent trajectories under a specified common continuation policy, and form empirical first-hit probabilities $\bar{F}_j$ for $j = 0, \dots, n$. With probability at least $1 - \delta$, simultaneously over all queries and horizons,*

$$|\bar{F}_j - F_j| \leq \sqrt{\frac{\log(2m(n+1)/\delta)}{2N}}.$$

If the critic differs from each empirical value by at most ϵ_{fit} , its score error on these queries is at most C_n times the sum of ϵ_{fit} and the displayed radius.

Proof. For each fixed query and horizon, the N hit indicators are independent bounded Bernoulli variables. Apply Hoeffding’s inequality (Hoeffding, 1963) and a union bound over $m(n+1)$ averages. Different horizons of the same trajectory may be dependent; the union bound does not require independence between those averages. The triangle inequality bounds critic error, and summing the $n+1$ errors with factor Δ proves the final claim. $\square$

This proposition gives a validation procedure; we do not report such an experiment here. The bound applies only to the tested policy and queries. It does not establish accuracy for optimal reachability or for arbitrary future CEM queries. Applying it would require the independent validation trajectories described above.

A.9 CONVERGENCE OF TABULAR EXPECTILE TD

Expectile regression (Newey & Powell, 1987) underlies implicit Q-learning (Kostrikov et al., 2022) and the HILP/GAS representation objective. We give the standard contraction proof for an unrestricted value table to explain what expectile TD guarantees and why that guarantee does not directly apply to neural TDR training.

For a bounded random variable Y and $\tau \in (0, 1)$, let $e_\tau(Y) = \arg \min_v \mathbb{E}[\ell_\tau(Y - v)]$. Strict convexity of the loss gives uniqueness. Its first-order characterization is

$$\tau \mathbb{E}[(Y - e_\tau(Y))_+] = (1 - \tau) \mathbb{E}[(e_\tau(Y) - Y)_+]. \quad (26)$$

The characterization also holds for degenerate Y .

Lemma 20 (Expectile monotonicity and stability). *If $Y \leq Z$ almost surely then $e_\tau(Y) \leq e_\tau(Z)$. For a constant c , $e_\tau(Y + c) = e_\tau(Y) + c$. Consequently, if $|Y - Z| \leq d$ almost surely, $|e_\tau(Y) - e_\tau(Z)| \leq d$.*

Proof. Define $h_Y(v) = \tau \mathbb{E}[(Y - v)_+] - (1 - \tau) \mathbb{E}[(v - Y)_+]$. It is continuous and strictly decreasing in v and crosses zero at $e_\tau(Y)$. When $Y \leq Z$, both positive-part terms imply $h_Y(v) \leq h_Z(v)$ for every v . Their zeros therefore have the same order. Adding c to both the random variable and the candidate value leaves every residual unchanged, proving the shift identity. Finally $Z - d \leq Y \leq Z + d$ and the first two claims give the stability bound. $\square$

For the next proposition only, use a finite deterministic MDP with absorbing goal, one-step reward -1 off the goal, discount $\gamma \in (0, 1)$, and fixed offline action probabilities $\mu(a \mid x)$ on nonempty supported action sets. On functions $U(g) = 0$, define

$$(\mathcal{T}_\tau U)(x) = \begin{cases} 0, & x = g, \\ e_\tau\{-1 + \gamma U(f(x, A)) : A \sim \mu(\cdot \mid x)\}, & x \neq g. \end{cases} \quad (27)$$

We fix the goal value at zero, including when a transition reaches the goal. The deterministic assumption matters: in a stochastic MDP, taking an upper expectile of sampled transitions can favor lucky random outcomes as well as better actions. It need not equal maximizing expected action values.

Proposition 21 (Tabular expectile convergence and residual bound). *$\mathcal{T}_\tau$ is a γ -contraction in sup norm with a unique fixed point U_τ . Its iterates satisfy*

$$\|U^{(k)} - U_\tau\|_\infty \leq \gamma^k \|U^{(0)} - U_\tau\|_\infty.$$

For any approximate value $\tilde{U}$ that is zero at the goal and satisfies $\|\tilde{U} - \mathcal{T}_\tau \tilde{U}\|_\infty \leq \epsilon_{\text{TD}}$,

$$\|\tilde{U} - U_\tau\|_\infty \leq \epsilon_{\text{TD}} / (1 - \gamma). \quad (28)$$

Proof. Use the same sampled action A in both backups. Their sampled targets differ by at most $\gamma \|U - W\|_\infty$. Theorem 20 gives the contraction, with identical zero values at the goal. The finite-dimensional space of functions that are zero at the goal is complete in sup norm, so the contraction

mapping theorem gives a unique fixed point and the iterate bound. For the residual claim, add and subtract $\mathcal{T}_\tau \tilde{U}$:

$$\|\tilde{U} - U_\tau\|_\infty \leq \epsilon_{\text{TD}} + \gamma \|\tilde{U} - U_\tau\|_\infty.$$

Rearrange using $1 - \gamma > 0$. $\square$

Proposition 22 (Finite-expectile bias). *Let U_* be the optimal discounted value restricted to the supported actions. Suppose some maximizing action at each nonterminal state has sampling probability at least $p_{\min} > 0$. If the span of the action targets $-1 + \gamma U_*(f(x, a))$ at each state is at most R_Q , then*

$$\|U_\tau - U_*\|_\infty \leq \frac{(1 - \tau)R_Q}{\tau p_{\min}(1 - \gamma)}. \quad (29)$$

Also $U_\tau \leq U_*$ pointwise.

Proof. Let M and m be the largest and smallest action targets at a state and let $v = e_\tau(Y) \in [m, M]$. A maximizing action has probability at least $p_{\min}$. By equation (26),

$$\tau p_{\min}(M - v) \leq \tau \mathbb{E}[(Y - v)_+] = (1 - \tau) \mathbb{E}[(v - Y)_+] \leq (1 - \tau)(M - m).$$

Thus the expectile backup of U_* differs from its maximum backup by at most $(1 - \tau)R_Q/(\tau p_{\min})$. Since the maximum backup of U_* is U_* , apply Theorem 21’s residual bound. For the ordering, $\mathcal{T}_\tau U \leq \mathcal{T}_* U$ and both operators are monotone. Starting expectile iteration from U_* gives a nonincreasing sequence. Its limit is U_τ by contraction, proving $U_\tau \leq U_*$. $\square$

Why this is not a neural convergence theorem. The actual TDR restricts U to negative symmetric Euclidean distances, uses two networks and lagged targets, and optimizes a sampled nonconvex loss. Its updates are not the exact unrestricted iterations in Theorem 21. Even a perfectly optimized distance model may not represent U_τ . If finite directed distances satisfy $d_*(x, g) \neq d_*(g, x)$, any symmetric approximation d obeys

$$\max\{|d(x, g) - d_*(x, g)|, |d(g, x) - d_*(g, x)|\} \geq \frac{1}{2}|d_*(x, g) - d_*(g, x)|.$$

Subtract the two true distances and apply the triangle inequality to the errors to obtain this bound. HILP discusses this limitation of symmetric representations (Park et al., 2024), and quasimetric learning treats directionality directly (Wang et al., 2023). High expectile, twin networks, or longer optimization do not remove it.

Even in the ideal deterministic shortest-path setting, a d_* -step path has discounted cost $(1 - \gamma^{d_*})/(1 - \gamma)$, rather than d_* . The discounted cost saturates with path length. Empirical median calibration u_k sets useful working scales but does not make the representation accurate everywhere. We therefore analyze the finite graph and local execution errors without assuming that neural TD learns an exact temporal metric.

A.10 A SEPARATION FROM TERMINAL L2 AND THE ROLE OF GEOMETRY

Proposition 23 (An exact-model L2 trap at any fixed planning horizon). *For every integer $K \geq 1$ and block duration $\Delta \geq 1$, there is a deterministic problem with an exact predictor, a connected graph, and equal-radius latent observations such that exact K -block terminal-L2 optimization keeps the agent at its start forever, even though the goal is reachable. The controller executes each full plan before replanning. Exact optimization of the graph-guided objective with its local switch and an exact reachability score succeeds in two calls, for every positive pair of normalization scales and $\beta \geq 0$.*

Proof. Use states $x_0, \dots, x_{2K} = g$. At each nonterminal state a block action either stays or advances one state; g is absorbing. A block lasts Δ primitive steps, with its state change at the last step. The budget is at least $2K\Delta$. Define injective raw latent observations on the unit circle by

$$z_g = (1, 0), \quad z_{x_i} = (\cos \theta_i, \sin \theta_i), \quad \theta_i = \frac{\pi}{3} + \frac{\pi i}{3K}, \quad 0 \leq i < 2K.$$

The start has squared L2 distance one from the goal. Every intermediate state has strictly larger distance, since $\pi/3 < \theta_i < \pi$ for $0 < i < 2K$ and $\|z_{x_i} - z_g\|^2 = 2 - 2\cos \theta_i$ increases on that

interval. From x_0 , the endpoints reachable within K blocks are precisely $x_0, \dots, x_K$; the goal is unavailable. The unique L2-minimizing endpoint is therefore x_0 , achieved by staying throughout the call. Replanning repeats the same choice indefinitely.

Set $\psi(x_i) = i$, $r_{\text{TD}} = 1$, and $\lambda = K$ in calibrated TDR units. Place singleton regular graph nodes at $x_0, \dots, x_{2K-1}$; the stated goal attachment rule connects g to x_{2K-1} . The radius graph has unit edges between successive states, so $D_g[x_i] = 2K - i$. At the start, either tied attachment x_0 or x_1 gives total path length $2K$, and its first λ -crossing target is x_K . An endpoint x_j , $0 \leq j \leq K$, has base cost $K - j$. Under the fastest continuation policy, its grid hitting time is $T_j = (2K - j)\Delta$. For every clock index $n \geq 0$, the exact survival score $\min(T_j, C_n)$ is nonincreasing in j . Thus the all-advance sequence ending at x_K strictly minimizes the base cost and weakly minimizes the critic cost. Shared centering and positive scaling preserve its strict combined-cost advantage for every $\beta \geq 0$.

At x_K , $d_\psi(x_K, g) = K = \lambda$, so the local L2 branch applies. A second all-advance sequence reaches g in K blocks and has both base cost and critic score zero. All other endpoints have strictly positive base cost and nonnegative critic score. Exact optimization therefore reaches the goal on this second call. $\square$

Taking $K = 5$ and $\Delta = 5$ gives the paper’s 25-step plan and execution cadence, with success in 50 steps. Appending zero coordinates embeds the construction in higher dimensions. We supply its radius and representation as fixed controller inputs; we do not claim that neural training recovers them. Both planners optimize their costs exactly in this example, so it isolates the effect of the objective. Finite-sample CEM may miss the best sequence, and the empirical tasks need not have this structure. The example therefore does not imply that every graph improves on L2.

Why a Gaussian shell is not a failure theorem. If z_1, z_2 are independent draws from $\mathcal{N}(\mu, \sigma^2 I_d)$, then

$$z_1 - z_2 \sim \mathcal{N}(0, 2\sigma^2 I_d), \quad \frac{1}{2}(z_1 + z_2) - \mu \sim \mathcal{N}(0, \frac{1}{2}\sigma^2 I_d).$$

Therefore independent-pair midpoint radii have the same distribution as a single centered radius divided by $\sqrt{2}$, and pairwise distances have the radius distribution multiplied by $\sqrt{2}$. This follows directly from sums of independent Gaussian vectors. It motivates the interpolation diagnostic, but says nothing about which action sequences a predictor can execute. On a sphere of radius R , two points separated by angle $\theta \in [0, \pi]$ have chord distance $2R \sin(\theta/2)$, strictly increasing with the shorter arc length $R\theta$. Thus chord and arc distances can give identical rankings. A shell alone does not imply a defective planning cost.

Similarly, a narrow distribution of start–goal L2 distances does not prove that small candidate-dependent changes in that distance contain no useful signal. The latent-ranking measurements examine that question on the selected candidates. The proposition gives a concrete failure mechanism: the available actions require the agent to increase L2 distance before it can reach the goal.

A.11 WHAT THE GUARANTEES ESTABLISH AND HOW THEY CAN FAIL

The combined theorem bounds the time to reach the goal under explicit assumptions. It does not prove convergence of a training algorithm. Each assumption identifies a way the controller can fail:

1. **A finite graph route is required.** If no eligible node connects to the goal, then $\Phi(z) = \infty$. Enlarging the goal radius adds a nearby attachment but does not ensure an executable connection.
2. **The selected target must be executable in one call.** The comparison sequence must end within η of the target. Median calibration does not guarantee this, and the first-crossing rule can select a target beyond the lookahead distance.
3. **Model-error bounds must cover both compared sequences.** Error under logged actions alone does not bound error on CEM’s selected actions. In stochastic systems one must also bound the relevant expected or conditional execution error.
4. **Search and critic tradeoffs must preserve tracking.** The quantities $s_b \epsilon_{\text{opt}}$ and $\beta(s_b/s_S)\omega$ both increase ρ . If the critic strongly favors plans that miss the target, the controller may fail to make graph progress.

5. **The local controller must stay in its region and reach the goal.** The limiting error bound must lie below the success tolerance, and sufficiently small latent error must imply the environment’s goal condition.

One can check these conditions on the implemented controller, but the reported tables do not estimate all of them at every required state. The theory therefore explains how the method can succeed without certifying each evaluated episode numerically. Poor target tracking, a conflicting critic, or both could explain the Cube ablations; the aggregate results do not distinguish these causes. The full method need not outperform TDR, graph-only planning, or terminal L2 in every environment.

Our contribution is to combine the algorithm’s restricted graph attachment, first-crossing targets, median/IQR normalization, full-plan execution, and repeated switching decision in the bounds equations (9) to (11). We use the standard tail-sum identity, Bellman nonexpansiveness, expectile contraction, and concentration inequalities to support that analysis. The policy-consistency gap and counterexamples show why their assumptions matter for this controller.

B PLANNING ALGORITHM

Algorithm 1 One full planning call of graph-guided latent MPC

Require: frozen E, P and history; TDR ψ ; graph centers y_v and $D_g[v]$; critic V ; budget B ; executed steps t ; $B - t \geq L$

- 1: $z_0 \leftarrow E(o_t)$; $h_{\text{rem}} \leftarrow \min(50, B - t - L)$; $n \leftarrow \lfloor h_{\text{rem}}/\Delta \rfloor$
- 2: **if** $d_\psi(z_0, z_g) \leq \lambda$ **then**
- 3: $b(\hat{z}) \leftarrow \|\hat{z} - z_g\|^2$
- 4: **else if** $A(z_0) \neq \emptyset$ **then**
- 5: Select v^* by equation (3)
- 6: Walk a shortest path, including the start attachment, to its first λ -crossing node (or the goal); denote its coordinate y_q
- 7: $b(\hat{z}) \leftarrow \|\psi(\hat{z}) - y_q\|$
- 8: **else**
- 9: Use goal L2 as a fallback; the graph-progress bound does not cover this case
- 10: **end if**
- 11: **for** 30 CEM iterations **do**
- 12: Sample 300 sequences of five action blocks and predict their endpoints
- 13: Compute b and $\text{ET} = \Delta \sum_{j=0}^n (1 - V_j)$
- 14: Normalize each score with its population median and floored IQR
- 15: Refit the CEM distribution on the 30 lowest combined-cost samples
- 16: **end for**
- 17: Execute the returned mean sequence for up to $L = 25$ steps; end the episode when the goal condition holds
- 18: Advance t by the number of steps actually executed

The planner keeps the target and remaining-time budget fixed during each call and recomputes normalization for each CEM population. The theory compares the returned mean sequence with a feasible comparison sequence using the final population’s constants. If the remaining budget cannot fit a full plan, a shorter plan must use its actual execution length when computing the time left afterward.

C EXTENDED COMPONENT ABLATIONS

Table 6 expands the Push-T ladder with the graph cost-to-go baseline, critic-weight variation, and alternative graph radii. The cost-to-go variant does not improve over direct TDR on these tasks. The best observed weight varies by protocol: among the four ET weights, $\beta = 2$ has the highest offset-50 mean and $\beta = 4$ the highest offset-100 mean, whereas $\beta = 1$ has the highest cross-episode mean. We use one shared weight, but its best value still depends on the task protocol. Changing the graph radius also changes coverage and possible subgoal execution error, both relevant to Theorem 1.

Table 6: Full Push-T ablation ladder (50 tasks, five seeds, mean $\pm$ reported standard deviation). A denotes graph subgoals with the local L2 switch. Radius sweeps are in TDR units.

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
L2 (LeWM)	91.2 $\pm$ 1.0	40.8 $\pm$ 3.7	13.2 $\pm$ 3.2	14.4 $\pm$ 3.9
TDR only	83.2 $\pm$ 4.1	48.4 $\pm$ 1.5	22.4 $\pm$ 5.3	14.0 $\pm$ 1.3
graph cost-to-go	83.2 $\pm$ 4.1	47.2 $\pm$ 1.6	21.6 $\pm$ 4.6	14.0 $\pm$ 1.3
graph subgoal, $\lambda = u_{25}$	80.8 $\pm$ 2.7	51.6 $\pm$ 2.3	22.4 $\pm$ 5.0	15.2 $\pm$ 3.0
+ ET, $\beta = 1$ (no switch)	86.0 $\pm$ 3.4	64.8 $\pm$ 3.0	43.6 $\pm$ 5.0	35.2 $\pm$ 2.0
+ local L2 switch (A)	88.0 $\pm$ 2.5	62.4 $\pm$ 2.7	30.8 $\pm$ 6.5	17.2 $\pm$ 2.7
A + $-\log V$, $\beta = 1$	77.6 $\pm$ 1.5	69.2 $\pm$ 2.4	47.2 $\pm$ 5.9	34.4 $\pm$ 7.3
A + ET, $\beta = 0.5$	88.4 $\pm$ 2.0	69.2 $\pm$ 2.7	43.2 $\pm$ 3.7	28.0 $\pm$ 3.8
A + ET, $\beta = 1$ (ours)	88.4 $\pm$ 2.3	70.0 $\pm$ 2.2	46.4 $\pm$ 3.2	42.4 $\pm$ 3.2
A + ET, $\beta = 2$	87.6 $\pm$ 3.2	72.4 $\pm$ 4.1	42.8 $\pm$ 4.1	35.2 $\pm$ 3.5
A + ET, $\beta = 4$	86.8 $\pm$ 2.0	65.6 $\pm$ 2.3	48.4 $\pm$ 5.0	38.4 $\pm$ 3.4
L2 + ET, $\beta = 1$	89.2 $\pm$ 3.0	58.4 $\pm$ 5.1	38.4 $\pm$ 7.1	31.6 $\pm$ 2.3
A + ET, L2 subgoal metric	86.0 $\pm$ 2.2	63.6 $\pm$ 4.8	45.2 $\pm$ 3.5	36.4 $\pm$ 2.9
A + ET, $\beta = 1$, $r_{TD} = 4$	82.8 $\pm$ 2.0	64.0 $\pm$ 1.3	48.4 $\pm$ 6.1	35.2 $\pm$ 2.7
A + ET, $\beta = 1$, $r_{TD} = 12$	89.6 $\pm$ 2.0	68.0 $\pm$ 3.8	48.0 $\pm$ 4.0	32.4 $\pm$ 6.1
A + ET, $\beta = 1$, $r_{TD} = 16$	91.2 $\pm$ 2.4	64.4 $\pm$ 6.6	47.6 $\pm$ 3.4	36.8 $\pm$ 4.5

C.1 CUBE

Table 7 uses Cube’s calibration $r_{TD} = 7.25$, $\lambda = 13.63$. Row (g) is a separate reevaluation of the full configuration and differs from the main result by at most 0.4 percentage points. The two tables therefore report separate measurements of the same configuration. TDR alone has the highest cross-episode mean, and the graph with a critic but no switch has the highest offset-100 mean. Adding the graph to TDR reduces the cross-episode mean from 36.8 to 28.4%. The final switch recovers 5.2 points at offset 25 but loses 7.6 at offset 100 before the critic is added. The components therefore rank differently on Cube and Push-T.

Table 7: Cube component ablation, success (%), mean $\pm$ reported standard deviation on 50 tasks and five seeds. Rows correspond to table 2; bold denotes the largest mean. Row (g) is a separate reevaluation and differs slightly from table 1.

	configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
	L2 (LeWM)	56.0 $\pm$ 3.8	34.4 $\pm$ 3.2	35.6 $\pm$ 1.5	16.4 $\pm$ 1.5
(a)	TDR only	50.0 $\pm$ 1.8	48.8 $\pm$ 2.0	50.4 $\pm$ 1.5	36.8 $\pm$ 4.3
(b)	graph subgoal	49.2 $\pm$ 3.0	46.8 $\pm$ 5.0	46.0 $\pm$ 3.6	28.4 $\pm$ 4.1
(c)	(b) + final switch	54.4 $\pm$ 2.9	44.8 $\pm$ 2.7	38.4 $\pm$ 3.7	28.4 $\pm$ 4.3
(d)	L2 + ET	64.0 $\pm$ 4.4	45.2 $\pm$ 2.4	47.6 $\pm$ 3.7	17.6 $\pm$ 3.4
(e)	(b) + ET	58.0 $\pm$ 1.8	46.8 $\pm$ 5.2	56.8 $\pm$ 5.2	28.4 $\pm$ 6.1
(f)	(c) + $-\log V$	56.8 $\pm$ 3.5	48.0 $\pm$ 2.2	39.6 $\pm$ 2.7	32.4 $\pm$ 4.8
(g)	(c) + ET (ours)	62.4 $\pm$ 5.4	49.2 $\pm$ 4.5	50.4 $\pm$ 3.2	27.2 $\pm$ 6.0
(h)	(g), L2 subgoal metric	63.6 $\pm$ 4.6	42.4 $\pm$ 2.9	47.2 $\pm$ 1.6	26.4 $\pm$ 4.6

D SUCCESS THRESHOLDS AND TIME BUDGETS

Table 3 reports Push-T cumulative success and threshold sensitivity. For cumulative percentages p_{j-1} and p_j , the fraction of previously unsolved tasks solved in window j , expressed as a percentage, is $100(p_j - p_{j-1}) / (100 - p_{j-1})$. We call this fraction the window hazard. Its denominator contains only tasks still unsolved at the window’s start, so changes can reflect which tasks remain. Our same-episode-100 window hazards after step 50 are 15.8, 17.2, and 15.7%; L2’s are 3.5, 0.5, and 0.9%. On cross-episode tasks, L2 reaches 6.4% in the second window and then falls below 1.5% in the

remaining windows. The later success gain is therefore concentrated in different budget windows for the two controllers.

Loosening thresholds preserves and sometimes widens the success gap. More runs satisfy the looser criteria. These totals do not show that every unsuccessful trajectory gets closer to its goal than the corresponding L2 trajectory.

E DEMONSTRATION INITIALIZATION

CEM initializes its mean to zero in action coordinates normalized with the training dataset’s action statistics. Zero in these coordinates is the dataset mean action, not necessarily physical inaction. The retrieval variant replaces this initial mean with a logged sequence. Let $\mathcal{T}$ contain training frames with at least $L = 25$ subsequent actions. For current latent z_0 and target y_{tar} , choose

$$i^* \in \arg \min_{i \in \mathcal{T}} \|\psi(z_i) - \psi(z_0)\| + \|\psi(z_{i+L}) - y_{\text{tar}}\|. \quad (30)$$

Use $y_{\text{tar}} = y_q$ in the graph phase and $\psi(z_g)$ in the local phase. Normalize the retrieved actions with the predictor’s training normalizer and reshape them into five blocks. We keep CEM variance, iterations, elites, cost, and execution cadence unchanged. Retrieval searches logged training segments. The planner does not use the held-out target actions from the diagnostic below.

Table 4 shows gains on Cube of 24.4, 9.6, 28.4, and 20.4 points at offsets 25, 50, 100, and cross-episode. Standard deviation decreases at each Cube protocol. On Push-T it loses 1.6–3.2 points on same-episode tasks and ties cross-episode; on Reacher it loses 2.0–14.0 points. We selected retrieval through a Cube configuration screen. Its results therefore describe an additional tuned variant and do not establish that this initialization rule improves every environment.

Action agreement. For 50 same-episode-25 tasks, we use a successful logged start-to-goal sequence as a reference. Other action sequences may also succeed. Table 8 compares it with retrieved and random training sequences. Retrieved actions have high agreement on Push-T, moderate agreement on Cube, and little agreement on Reacher. Under the unit-covariance Gaussian initialization, the log-density change for reference action vector a and retrieved mean m is $a^\top m - \|m\|^2/2$. Thus a negative density change means that retrieval makes that particular logged sequence less likely under the initial CEM distribution.

Table 8: Retrieved-action agreement with a successful logged action sequence on 50 same-episode-25 tasks. r is Pearson correlation of normalized actions; MSE is action mean squared error. $\Delta \log P$ is the median log-density gain for the logged sequence under a unit-covariance Gaussian initialized by retrieval. Win is the fraction closer to that sequence than zero initialization.

Env	r (retrieved)	r (random)	MSE (retrieved)	MSE (random)	$\Delta \log P$ / win
Push-T	0.90	−0.07	0.20	2.19	+16.6 / 98%
Cube	0.67	0.01	0.48	1.78	+19.6 / 88%
Reacher	0.04	0.04	1.87	1.93	−21.2 / 0%

These measurements suggest that retrieval helps CEM sample useful grasp-and-carry actions on Cube but provides less useful proposals on Reacher. Low agreement with one logged sequence does not rule out other successful actions. Likewise, high agreement on Push-T need not improve success if the original initialization already samples useful candidates.

F PREDICTOR ERROR, LATENT GEOMETRY, AND CANDIDATE RANKING

F.1 ROLLOUT ERROR AND TASK DISTANCE

Figure 3 and table 9 compare predicted endpoints under logged actions with true encoded endpoints. We sample the 3,000 starts per horizon in the same way for both measurements. Relative error is a ratio of reported medians, not a median of per-example ratios. The error increases from 1.00 at horizon 5 to 7.80 at horizon 100; relative error first falls from 18.9 to 14.1% at horizon 25 and then

rises to 39.8%. Relative error therefore does not increase monotonically across all horizons. These measurements under logged actions do not bound model error on CEM’s selected sequences.

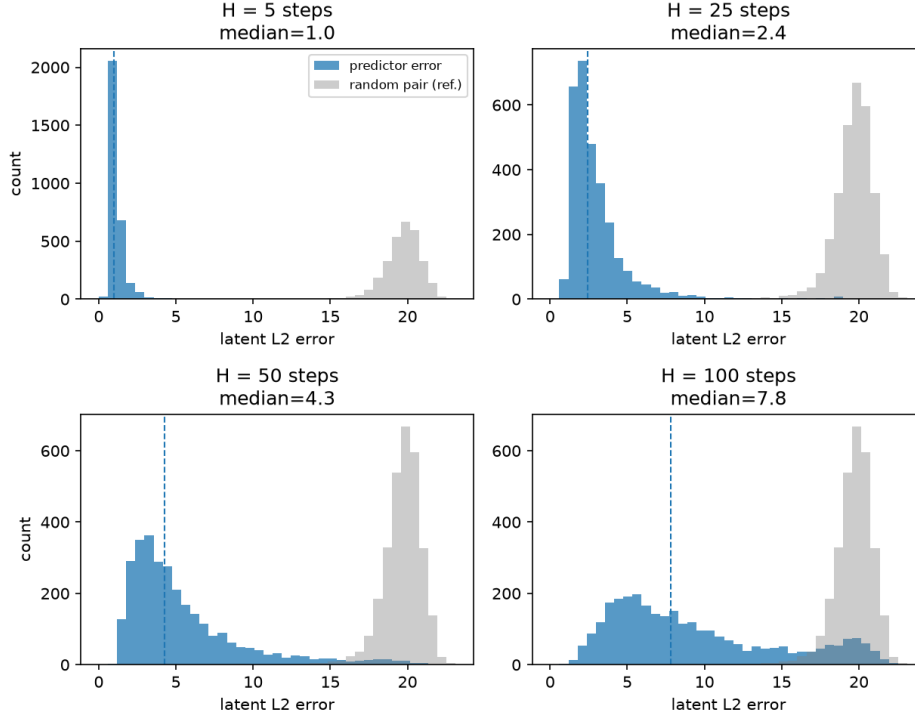


Figure 3: Push-T latent rollout error under logged actions as a function of horizon, with the random-pair latent-distance reference. We evaluate predictions under logged actions; the errors need not match those on CEM’s selected actions.

Table 9: Push-T: predictor latent rollout error vs. the true same-episode start→goal distance at the same horizon (both $n = 3,000/\text{horizon}$, identically sampled), median $\pm$ std in latent L2. Random-pair reference: mean 19.6 (theory: $\sqrt{2} \times 13.88 \approx 19.63$, section F.2). Relative error is the predictor-error median divided by the start→goal-distance median.

	$H = 5$	25	50	100
predictor error	1.00 ± 0.56	2.41 ± 2.09	4.28 ± 3.82	7.80 ± 5.35
start→goal distance	5.29 ± 2.76	17.06 ± 4.19	19.45 ± 3.29	19.60 ± 2.43
relative error (%)	18.9	14.1	22.0	39.8

Planning and execution length. Table 10 varies the CEM horizon and executed chunk together. Five-step plans perform poorly, consistent with a short lookahead missing useful multi-step approaches. A 100-step commitment also lowers success. The 25-step configuration performs best in this sweep, which does not cover all possible MPC settings. The 100-step cross-episode setting requires a final budget-limited segment. The main comparison uses only full 25-step chunks, so that boundary does not arise there.

F.2 GAUSSIAN-SHELL DIAGNOSTIC

An isotropic Gaussian fit to the 2,336,736 Push-T latents uses empirical mean μ and variance $\sigma^2 = \text{tr}(\Sigma)/192$. Its predicted centered-radius mean and the measured mean are both 13.88. The covariance participation ratio is $\text{tr}(\Sigma)^2 / \text{tr}(\Sigma^2) = 188.5$. The variance therefore spans many latent

Table 10: Push-T L2 success means (%) for different plan/execution lengths on 50 tasks and five seeds. Planning horizon equals the executed chunk; the main setting is 25. The comparison spans 5-, 25-, and 100-step commitments. Dashes denote settings not evaluated for the corresponding short budget.

Protocol (budget)	5 steps	25 steps	100 steps
same-episode 25 (50)	64.0	91.2	–
same-episode 50 (100)	19.2	40.8	–
same-episode 100 (200)	1.2	13.2	5.6
cross-episode (250)	4.4	14.4	4.8

directions. This statistic does not establish a joint Gaussian distribution or independence between same-episode endpoints. Figure 4 compares the full radius histogram with the fit; the midpoint comparison appears in the main text (figure 2).

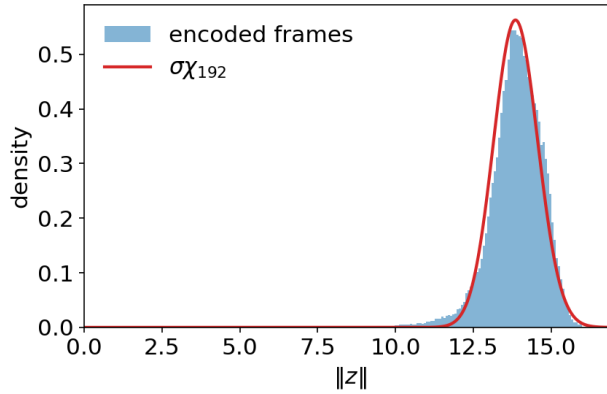


Figure 4: Push-T centered radii of encoded frames and the fitted isotropic Gaussian reference. Concentrated radii alone do not show that the full latent distribution is Gaussian or determine which trajectories the action-space planner will choose.

For 200,000 pairs at offset 100, reported mean endpoint radii are 14.15 and 13.60; the actual logged midpoint has mean radius 13.81. The arithmetic midpoint has mean radius 10.03, and 96.4% of these interpolations lie below the fitted shell’s first-percentile radius. Straight-line interpolation therefore places most of these midpoints inside the observed shell, away from the usual data radii. The measurement does not show that a planner optimizing terminal L2 follows those straight lines. Section A.10 explains why shell geometry alone is insufficient to prove a planning failure.

F.3 CANDIDATE-RANKING DIAGNOSTIC

For 25 Push-T tasks per same-episode protocol, the first CEM iteration begins from the same unadapted population of 300 candidates. The proposed objective selects 30 elites. We execute those candidates in the simulator and compare both costs with a recovery-time diagnostic that uses simulator state information. Both costs therefore use the same analyzed candidates, but the proposed objective selects that subset from the initial population. The correlations in table 11 favor L2 at offset 25 and the proposed score at offsets 50 and 100. Only 11 of 25 offset-100 tasks have non-constant recovery labels because of censoring. We report descriptive correlations on these selected candidates. We do not test statistical significance or assume that censoring is independent of task difficulty. The correlations suggest better temporal ranking at longer horizons within this subset.

Table 11: Mean Spearman correlation between candidate cost and simulator recovery time for the 30 elites selected by the proposed method’s first CEM iteration. Of 25 tasks, 25, 24, and 11 have nonconstant diagnostic labels at offsets 25, 50, and 100. We report descriptive correlations without a significance test.

protocol	$\rho(\text{OUR})$	$\rho(\text{L2})$
same-ep 25	0.327	0.396
same-ep 50	0.203	0.086
same-ep 100	0.191	−0.064

G COMPUTE AND REPORTED TRAINING SETTINGS

Table 5 separates shared encoding, per-seed preparation, and online decision cost. Summing the listed preparation stages gives 190.8 minutes including encoding, or 141.2 minutes after shared encoding. The planning ratio is $1.80/1.14 \approx 1.58$. These timings describe one device and can change on other hardware. Table 12 reports component counts. We train both TDR networks and use only the first during planning.

Table 12: Reported component parameter counts on Push-T. We sum the listed components to obtain the frozen subtotal. Rounding accounts for differences between per-member and ensemble totals.

LeWM encoder (ViT-tiny)	5.50M
LeWM predictor	10.79M
LeWM action encoder	0.16M
Sum of the three listed components	16.45M
TDR (used, ψ_0 only)	0.64M
TDR (full 2-net ensemble)	1.29M
Critic (per member)	0.83M
Critic (5-member ensemble)	4.13M

G.1 HYPERPARAMETERS

Tables 13 and 14 list the reported network, optimizer, graph, and planning settings. The critic maps logits to probability outputs. We express the integer temporal-efficiency lag k_{TE} in environment steps and the graph radius in calibrated TDR units. The learned components use the shared settings. We set the graph radius and lookahead from each environment’s training-data calibration.

H RENDERING AND ACTION NORMALIZATION

To check rendering, we render three dataset states again and compare them with their stored images, requiring mean pixel error within $4/255$. Reacher rendering with MuJoCo 3.12 instead of the dataset’s reported 3.5 version changes the floor texture, with reported average pixel discrepancy $18/255$ and median latent shift 15. The check gives discrepancies of $0.5/255$ on Push-T and at most $0.2/255$ on Cube. This check tests agreement with stored observations; larger changes in renderer settings could still affect the encoded latents. Predictor inputs and retrieved action proposals use the full training dataset’s action mean and standard deviation.

I COMPARISON WITH PUBLISHED EVALUATIONS

Panel (B) of table 1 reproduces short-horizon results reported by LeWM (Maes et al., 2026) on its own task sets. Those results use a different evaluation protocol, so they do not provide paired comparisons on our fixed tasks. SAGE reports Push-T success of 64.7% at offset 150 versus 12.7% for its LeWM baseline (Cheng et al., 2026). We do not evaluate that offset in the main table. Our matched comparison establishes improvement over LeWM on the evaluated tasks. It does not establish superiority over every published planner.

Table 13: Shared training and planning settings. The critic maps its final scalar logits to probabilities in $[0, 1]$. We set the radius and lookahead using each environment’s calibration in table 14.

component	hyperparameter	value
TDR ψ	architecture	3×512 , LN+GELU
	embedding dim	32
	expectile τ	0.99
	discount γ	0.99
	training steps	50,000
	batch size	1024
	learning rate	3×10^{-4}
Graph	TE threshold θ_{TE}	0.9
	clustering radius	$r_{TD}/2$
	goal attachment radius	$\max(r_{TD}, 1.2 d_{\min})$
Critic V	ensemble members	5
	architecture (per member)	3×512 , SiLU
	max horizon $h_{\max}$	50 env steps
	horizon grid	multiples of 5, 0–50
	training steps	60,000
	Bellman pessimism κ	1.0
	loss weights ($\lambda_{\text{Bellman}}, \lambda_{\text{mono}}$)	1.0, 1.0
	learning rate	3×10^{-4}
	EMA target τ	0.995
CEM	population / iterations / elites	300 / 30 / 30
	horizon K (blocks $\times$ frameskip)	$5 \times 5 = 25$ env steps
	replan interval (receding horizon)	25 env steps
Criterion (equation (7))	critic weight β	1
	critic cost	ET from equation (6)
	final-phase metric	squared latent L2

Table 14: Environment-specific calibrated TDR scales. u_k is the median distance between training frames k steps apart; $\lambda = u_{25}$ and $r_{TD} = u_{12}$. The remaining settings listed in table 13 are shared.

	Push-T	Reacher	Cube
r_{TD} (TDR units)	7.59	3.72	7.25
$\lambda = u_{25}$ (TDR units)	13.53	5.83	13.63